

# Choupo

The Open Source Chemical Process Simulator

# EduTools Guide

Choupo-dev

Copyright © 2026 Vítor Geraldes.

Authors: Vítor Geraldes.

Curators: Mónica Faria, Departamento de Engenharia Química, Instituto Superior Técnico — the batch UF/NF membrane tool.

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# 1 What this guide is — and what it is not

EduTools is the plane of Choupo's browser application where the *classical graphical methods* are constructed, most of them over curves the engine computes: the McCabe–Thiele staircase, the Kremser recovery curve, the $\varepsilon$–NTU chart, the Levenspiel plot, the Van Heerden ignition diagram, the pinch composite curves, and the rest. Most tools draw a construction a student has met on paper on numbers a real solver produced, and put the method's answer beside the engine's; some work a closed form on the page, quote a cited source or a recorded run, or start from values the reader sets. Every result on a page says which: it carries one of four origin marks — *engine*, *page*, *reference* or *assumption* — explained in a legend above the tool.

> **Key idea.** This guide is a **lesson companion**, not a theory manual. It says what each construction is *for*, what to turn, what to watch, and what the picture is silent about. It contains no derivations at all.

That absence is deliberate and it is architectural. Every equation behind these tools already has exactly one home — `docs/theoryGuide.pdf` — and every live tool declares, in its registry entry, the named destination in that guide where its derivation lives. A second document carrying the same derivations would be a second home for the same statement, and within one semester the two would disagree. So this guide *points*, and the Theory Guide *derives*. When you want to know why $\varepsilon = (1 - e)/(1 - C_r e)$, follow the pointer. When you want to know what a class will get wrong about it, you are in the right document.

## The five questions each tool section answers

Every tool documented here is written to the same five headings, in the same order, because a lesson is more portable than a description:

1. **The question** — in one sentence a student would recognise from a problem sheet.
2. **What to vary, what to watch** — the specific knob that makes the lesson visible, and what should happen when you turn it.
3. **The misconception the construction exposes** — what students actually get wrong, not merely what is true. This is the part worth reading twice.
4. **What the picture does not say** — the limits of the construction. In this project that is treated as part of teaching it, never as an apology for it.
5. **Where the derivation lives** — a pointer into the Theory Guide, never a copy of it.

## Where the other material is

- **Theory Guide** — every derivation, every hypothesis, every worked example. The authority for all physics.
- **Explorer Guide** — the *property surfaces* ($T$-$x$-$y$, $\gamma(x)$, $P^{\mathrm{sat}}$ scans). The split criterion between the two planes is stated in the application itself: method-construction → EduTools, property-surface → Explorer.
- **Tutorials Guide** — the runnable cases. Every EduTools tool runs one of them; this guide names which.
- **User Guide** — authoring cases on disk, which is where any finding from a tool should end up.

# 2 How a tool feeds itself

> *What you'll learn.* Where the numbers on the screen come from, why a tool works with no case open, and what a knob is actually allowed to change.

## 2.1 Opening a tool

EduTools is a tab of its own. The tool chooser leads the chrome row on that tab; on a case tab showing EduTools in its body the chooser is one slim row above the construction. Either way the list is the same, and a construction that is on the roadmap but not yet built is shown *disabled* rather than hidden — the menu states the roadmap instead of implying it.

Every tool is deep-linkable:

```
?workspace=methods&tool=<id>
```

The `methods` key is a contract and does not change with the workspace's caption, so a link already in a student's notes goes on working. The tool identifiers are the ones listed in §3 and §7.

Under the chooser sits a one-line caption of what the construction teaches, and beside it a book icon that opens the Theory Guide *at this tool's own section*. That icon is the pointer this whole guide is built around: a tool that cannot name its derivation is not finished, and a test in the application refuses an anchor the Theory Guide does not define.

## 2.2 Paths, and finding a tool

Above the chooser sits one compact row, *Path*. A path is a reading order over tools that already exist — three ship: distillation, thermodynamics and reaction engineering — and choosing one shows its steps as a numbered strip with the current step marked, a sentence saying what that step adds, and Previous and Next buttons. A path adds no page and no panel: it only moves between the same tools, and its address,

```
?workspace=methods&tool=<id>&path=<path id>
```

carries both, so a teacher can send a student the third step of a path as one link. Each tool also has a one-line blurb of at most twelve words, which is what the public landing page prints large on its cards (the longer description sits below it). The landing page also has a text filter over the tools, and its discipline shelves fold; without JavaScript it is simply the full list.

## 2.3 Classroom, or your own run

Most tools default to **Classroom**: the tool clones a bundled tutorial — its *witness* case — writes your knob values into that case's own dictionary files, and solves it in the browser with the WebAssembly build of the same solver the command line runs. Nothing needs to be open, and nothing on disk is touched.

When the application's current run also carries the surfaces a tool reads, a source toggle offers **Current run** beside the classroom witness. A tool that a run cannot feed says so and stays empty; it never draws something else wearing the same name.

> **Key idea.** A knob turns a *declared scalar* in the witness's dictionary and nothing else. The engine then recomputes everything. A knob that cannot find its key fails loudly rather than silently doing nothing, and a knob that would not reach the physics is deliberately absent — each absence recorded with its reason in the tool's own source.

That last rule has teeth. Several plausible-looking controls are missing on purpose: a pressure that a liquid-phase rate law never reads would be a placebo control; an integrator step count sitting beside physical knobs invites reading numerics as physics; a kinetic pre-exponential that a witness deliberately re-tuned would leave the tool teaching a case other than the one it names.

## 2.4 Why the picture can be trusted

Three standing rules govern this plane, and they are worth telling a class about, because they are the reason the construction and the simulator cannot quietly diverge.

1. **No physics in the browser.** Every curve is an engine column, an engine KPI, or an engine-written report file. The browser draws.

2. **Method geometry is the one exception, and it is labelled.** A staircase, a trapezoid area, a textbook closed form, a heat cascade re-run over published curves — these *are* the method being taught, so a few tools own them. Where that happens the tool says so on screen, and the engine's own number is the judge: agreement is verification, and a disagreement is a finding about the drawing, never a moved answer.
3. **An absence stays an absence.** A number the engine did not publish is not rendered as zero and not invented. Solver advisories are lifted out of the run log and shown, because these tools have no log tab to bury them in.

**Common pitfall.** A tool that shows nothing is usually telling you something. "This run publishes no Van Heerden surface" means the reactor you solved was isothermal — there is no generation-versus-removal diagram for it. The honest empty state is a diagnosis; read it before changing the tool.

# 3 The constructions

The constructions documented in full follow. Several further live tools are listed in §7 with what is missing from each, because a thin lesson is worse than an admitted gap.

The six were chosen to span the three transfer subjects a course actually has to cover, two constructions each, and to prefer the tools whose *misconception* is both common and hard to dislodge by telling:

- **Mass transfer** — McCabe–Thiele (§3.1) and Kremser (§3.3).
- **Heat transfer** — $\varepsilon$–NTU (§3.4) and pinch composite curves (§3.5).
- **Reaction** — Levenspiel (§3.6) and Van Heerden (§3.7).

## 3.1 Distillation: the McCabe–Thiele staircase

*Declared anchor:* `ch:distillation`.

### The question

*A binary feed of composition $x_F$ is to be split into a distillate $x_D$ and a bottoms $x_W$. At reflux ratio $R$ and feed quality $q$, how many equilibrium stages does it take, and on which stage does the feed enter?*

### What to vary, what to watch

> **Try it now.** Set a sharp split ($x_D = 0.95$, $x_W = 0.05$, $x_F = 0.5$), then drag $R$ *down* toward the red $R_{\min}$ tick on the slider. Watch three things at once: the staircase steps multiply, the *approach to pinch* bar fills and reddens, and at $R \le R_{\min}$ the construction stops with a banner naming the pinch abscissa. Then drag $R$ up and watch how little the stage count changes once you are clear of the cliff.

- $R$ — the primary knob. The slider can be bound to $R$ directly or to $R/R_{\min}$; the second binding is the design lens ("1.3× minimum") and makes the cliff a fixed landmark at 1.0 instead of a number that moves.
- $q$ — rotates the $q$-line about $(x_F, x_F)$. Watch the feed-stage index move, and watch $R_{\min}$ itself change: the same column, the same specifications, a different minimum reflux, because the feed arrived hotter or colder.
- $E_{MV}$ — the Murphree tray efficiency. Below 1 the staircase steps to a dashed *pseudo*-curve and the count becomes a count of real trays.
- **The partial-condenser switch** — adds exactly one equilibrium stage, and the readout says so.
- **The components, $\gamma$ model and $P$** — these are the only settings that re-run the engine, because they are the only ones that change the equilibrium curve. Everything else redraws instantly.

### The misconception the construction exposes

> **Common pitfall. That the equilibrium curve responds to the reflux ratio.** It is the commonest confusion in the whole subject, and it is usually invisible because on paper the student redraws the entire figure each time. Here the curve is a frozen engine result and the staircase is pure geometry over it, so turning $R$ at sixty frames a second while $y^*(x)$ does not so much as flicker makes the point physically: *reflux is an operating decision; the equilibrium curve is a statement about the mixture.* Change the pressure or the activity model and the engine re-runs and the curve moves — which is the same lesson from the other side.

Two further confusions the same picture settles:

- **That $R_{\min}$ is a property of the column.** It is the intersection of the $q$-line with the equilibrium curve — feed composition, feed quality, product specifications and thermodynamics, and no equipment at all. Turn $q$ and $R_{\min}$ moves although nothing has been built or unbuilt.
- **That more reflux is better because it needs fewer stages.** The chart shows only one half of that trade. Stage count falls steeply just above $R_{\min}$ and then flattens; the vapour traffic $V = (R+1)D$, and with it the reboiler duty, the condenser duty and the column diameter, goes on rising linearly and is *nowhere on this diagram.* A student who optimises on the staircase alone will choose a column that cannot be paid for.

### What the picture does not say

- **The operating lines are straight because Constant Molar Overflow is assumed.** The tool states this permanently under the chart rather than in a footnote. CMO is an energy assumption (equimolar latent heats),

and it is the one thing on the diagram that is not the real model. The equilibrium curve, by contrast, is the real curve at that pressure — not a constant-$\alpha$ toy.
- **No energy, no hydraulics, no diameter.** There is no reboiler duty, no condenser duty, no pressure drop, no flooding check. A column that is impossible to build converges perfectly well here.
- **Stages are not trays** unless $E_{MV} < 1$, and $E_{MV}$ is a *declared* number — from a correlation, a chart, or plant data. The engine will not invent it.
- **The azeotrope you see belongs to the declared model.** If no curated binary-interaction pair covers the system, the tool announces that it is drawing *ideal* mixing and that the diagram therefore cannot show an azeotrope at all. Read that banner: an absent azeotrope may be a fact about the mixture or a fact about the data coverage, and only the banner tells you which.
- **Binary only.** A third component has no axis here.

### Where the derivation lives

**Derivation.** Theory Guide, *Distillation column, Wang-Henke bubble-point* (`ch:distillation`) — the stage model, CMO, and the rigorous column this construction is the hand method for. The closed forms for the two limits the staircase locates geometrically, $N_{\min}$ and $R_{\min}$, are derived in *Shortcut distillation: Fenske–Underwood–Gilliland* (`ch:fug`). The Murphree efficiency and its effective-$K$ treatment are in *Tray hydraulics* (`ch:hydraulics`). The staircase construction itself — horizontal to the operating line, vertical to the equilibrium curve — is drawn and explained in the absorber chapter (`ch:absorber`), where the same geometry serves a counter-current cascade.

## 3.2 Equilibrium stages or a rate-based model?

*Declared anchor:* `ch:ratebased`.

*Tool:* `tool=equilibrium-or-rate-based`. *Cases it runs:* `column11_murphree` as shipped and with four declared numbers rewritten, and the three tray-efficiency witnesses `trayEfficiency01_yanagi14`, `trayEfficiency05_chanFair_yanagi14` and `trayEfficiency06_syedaChuang_yanagi14` (all in `tutorials/steady/distillation/`).

### The question

*I have to design or rate a column: is an equilibrium-stage model with one efficiency enough, or do I need a rate-based model, and how wrong can the simple one be?*

The page states first what the engine has: a declared Murphree efficiency on the Wang–Henke column, a computed tray efficiency for a binary at total reflux, and no rate-based column. Every recommendation on the page is either demonstrated by an engine run on the page or read in a paper the page lists, and the closing decision table names the origin of every row.

### What to vary, what to watch

**Try it now.** Demonstration 1 runs one benzene–toluene column at seven efficiencies, from 0.4 to 1.0, twice: as the tutorial ships it (a few per cent of toluene in the distillate) and taller at higher reflux (parts per million). Going from an efficiency of 0.7 to 0.6 multiplies the first impurity by about 1.2 and the second by about 4. Then count the trays a 10 ppm target needs at three efficiencies: 27 real trays at 1.0, 37 at 0.7, 63 at 0.4. Demonstration 2 runs three published correlations on one measured sieve tray: at an F-factor of 1.5 they span about 28 points, while the measured efficiency stays between 63 and 74 % over the normal range of load.

**Common pitfall. That a column built from equilibrium stages and one efficiency is safe because the separation is easy.** On a trace specification the efficiency acts once per tray and the error compounds; the uncertain number is the efficiency itself, which correlations for the same tray do not agree on.

### What the page takes from references

Four open conference papers, each read for the page, and four papers read by the project's coordinating session and not opened for the page: two commercial $H_2S$ strippers where one assumed efficiency put the bottoms impurity two orders of magnitude below the plant and the computed component efficiencies varied from about 20 to 60 % along the column (Taylor, Wilkinson and Kooijman, 2010); a quaternary azeotropic mixture whose measured trajectories crossed a distillation boundary that an equilibrium-stage model cannot cross (Springer, van der Molen and Krishna, 2002); a packed column at total reflux, near the boundary of the 2-propanol and water azeotrope, whose reboiler filled with water where the equilibrium-stage model predicted 2-propanol, so that model got the

product wrong, not a percentage (Pelkonen, Górak, Ohligschläger and Kaesemann, 2001); reactive absorption in sour gas treatment, where an equilibrium model got the $H_2S$ over $CO_2$ selectivity wrong (Thiele et al., 2006); and an industrial dehexaniser study on how efficiencies vary by component and tray (Jakobsson et al., 2002). From the three further papers read by the coordinating session the page takes: that with three or more components the component efficiencies are almost always unequal and can exceed one or fall below zero (Krishnamurthy and Taylor, 1985; Taylor and Krishna, 2000); that for a reactive column equilibrium stages serve preliminary design while final design, control and the dynamics call for the nonequilibrium model (Taylor and Krishna, 2000); and, in the costs, what a rate-based model needs (hardware data, mass-transfer and hydrodynamic parameters, diffusion coefficients, viscosity, surface tension), that the tray flow model alone moves the transfer rate by up to about 20 %, and the one paper's own count of the size of the problem (70 against 160 equations for ten stages and three components, 550 against no fewer than 1 300 for fifty stages and five), which its authors call a considerable difference but a minor drawback (Krishnamurthy and Taylor, 1985; Taylor and Kooijman, 2014). The trace specification is the page's own finding from demonstration 1: none of the papers names it as a reason to choose a rate-based model.

### What this tool does not carry

No rate-based model; no component efficiencies that differ; demonstration 1 is a near-ideal binary on constant molar overflow, with the trace column's numbers and the targets chosen by the page. The measured tray efficiencies are the tree's transcription, not re-read for the page. A review often cited for these recommendations (Real-world modeling of distillation, 2003) could not be opened, so the list of situations is incomplete. The measured tables and the dropped candidates are in `docs/design/equilibrium-or-rate-based.md`.

### Where the derivation lives

**Derivation.** Theory Guide, *Columns from their internals: total reflux, computed tray efficiency, packed beds* (`ch:ratebased`) for the computed efficiency, and *Distillation column, Wang-Henke bubble-point* (`ch:distillation`) for the column and its declared Murphree efficiency.

## 3.3 Absorption: the Kremser shortcut, judged

*Declared anchor:* `ch:absorber`.

### The question

*A gas carrying a soluble solute is washed counter-currently with a lean solvent. How much of the solute do $N$ stages recover — and does the one-line shortcut agree with what the column actually does?*

### What to vary, what to watch

The construction has two halves and the lesson is in the gap between them. The curve is the Kremser closed form, recovery against $N$ at the absorption factor $A = L/(KV)$ the engine published. The point is the engine's own stagewise recovery for the column that was solved. The chip between them is the labelled deviation.

> **Try it now.** Start from the witness (an ammonia/water absorber, six stages, both feeds at 298.15 K). Now raise the ammonia in the gas feed, step by step, and watch *two* chips together: `dT_rise` climbing as more heat of absorption is released, and the deviation between the shortcut and the column growing with it. Then put the ammonia back and raise the solvent flow instead: $A$ rises, recovery rises, and the deviation shrinks.

- **Stages** $N$ — the knob students reach for first. Watch how quickly the curve flattens: past a handful of stages, more stages buy very little.
- **Solvent flow** $L$ — the knob that actually moves the answer, because it moves $A$.
- **Solute feed flow** — the knob that breaks the shortcut, by making the column non-isothermal.
- **Feed temperatures** — both of them, gas and solvent, with the `nonIsothermal` flag and `dT_rise` as the readout.

### The misconception the construction exposes

> **Common pitfall. That the deviation is an error.** A student who has only met the closed form reads any gap between shortcut and simulator as a defect — a bad $K$, a loose tolerance, a bug. It is neither. Kremser assumes *one constant* $A$, which means a straight operating line and a straight equilibrium line, which means an isothermal column. Absorption is exothermic. The solvent heats as it descends, $K$ rises with temperature, and $A = L/(KV)$ therefore *falls* exactly where the column is doing its work. The engine re-evaluates $K$ stage

by stage; the shortcut cannot. The gap is the assumption failing, and it is real.

The direction matters as much as the size, and it is the part worth labouring: for a soluble, exothermic solute the isothermal Kremser is *optimistic*. It promises recovery the column will not deliver. A shortcut that erred the other way would merely be conservative; this one sends you to site with a column that underperforms.

A second confusion, milder but persistent:

- **That $A > 1$ settles the matter.** $A$ is a ratio of two slopes, not a pass mark. At $A$ barely above one, recovery approaches its ceiling so slowly that no realistic stage count gets there; at $A$ comfortably above one, a few stages do almost everything. The curve makes the difference between those two situations obvious in a way that the inequality $A > 1$ never can.

### What the picture does not say

- **The $A$ on the chip is a single number for a column whose $K$ varies.** It is evaluated at the feed temperature. That is precisely why the two sides of the deviation chip are not the same calculation — and precisely what makes the chip informative.
- **Absorbers only.** A stripper publishes a stripping factor $S = KV/L$ and a temperature *drop*, not these keys, so it does not activate this tool. That is a correct refusal, not a missing feature: the mirrored construction is a different lesson.
- **Stages are theoretical.** No packing height, no HETP, no tray efficiency, no flooding.
- **No solvent circuit.** The lean solvent arrives lean. A real plant regenerates it, and the regenerator's residual solute sets a floor on the recovery that no stage count can beat. That loop is not on this diagram.

### Where the derivation lives

**Derivation.** Theory Guide, *Absorber and stripper: the counter-current cascade* (`ch:absorber`). The absorption factor $A_i = L/(K_i V)$ and the tridiagonal stage system are in *The counter-current cascade and the absorption factor*; the reason the engine does not use the closed form, and the temperature bulge that makes the isothermal shortcut optimistic, are in *Why the energy balance matters*.

## 3.4 Heat exchange: effectiveness against NTU

*Declared anchor:* `ch:hx-entu`.

### The question

*I have an exchanger of area $A$ and overall coefficient $U$, and two streams with known inlet temperatures and flows. What duty do I actually get — and if I want more, will more area give it to me?*

### What to vary, what to watch

The chart is the family of $\varepsilon$(NTU) curves at several capacity ratios $C_r$, with the solved exchanger's own point placed on the curve for its own $C_r$. A chip reports whether the drawn geometry reproduces the engine's effectiveness; when it does not, that is a statement about the configuration the tool assumed, never about the point.

**Try it now.** Double the area, from 10 to 20 m$^2$, and read the duty before and after. Then double it again, to 40. Note how much of the first doubling you got back on the second. Now set the two heat-capacity rates far apart (raise one flow, drop the other), so $C_r \to 0$, and switch between counter-current and co-current: the two curves close up until they lie on top of each other.

- **Area $A$ and coefficient $U$** — both enter only through NTU $= UA/C_{\min}$, which is the first thing to notice: a better coefficient and a bigger exchanger are the same move on this chart.
- **The two flows** — they set $C_{\min}$, $C_{\max}$ and hence both NTU and $C_r$, so a flow change slides the point along one curve *and* moves it to another.
- **The two inlet temperatures** — they do not move the point at all. They set $Q_{\max}$, and therefore the duty in kilowatts, but $\varepsilon$, NTU and $C_r$ are all independent of them. Verifying that is a good five minutes.
- **The configuration** — counter-current, co-current, or one shell with two tube passes.

### The misconception the construction exposes

> **Common pitfall. That duty scales with area.** NTU is linear in $A$; $\varepsilon$ is emphatically not. Every curve on the chart bends over, and past roughly NTU = 3 the return on further area is nearly nothing — the exchanger is already delivering most of what the inlet temperatures permit, and no amount of metal buys what thermodynamics has capped. Students who have only seen $Q = UA\,\Delta T_{LM}$ read a proportionality that is not there, because $\Delta T_{LM}$ collapses as the exchanger gets good.

The second one is more interesting, and the chart proves it in one move:

- **That counter-current is always better.** It is never worse, but at $C_r = 0$ — one stream condensing or boiling, or simply so large that its temperature barely moves — the two arrangements give *identical* effectiveness, and the curves coincide exactly. The advantage of counter-current flow is purchased entirely by the temperature change of the smaller stream, so when that change is absent there is nothing to buy. This is worth doing on the screen because it is the kind of fact that is remembered as a slogan and applied where it does not hold.
- **That $\varepsilon$ is an efficiency, so $\varepsilon = 0.75$ means a quarter is wasted.** Nothing is wasted; the exchanger is adiabatic to its surroundings and every joule that leaves the hot stream arrives in the cold one. $\varepsilon$ is normalised by $Q_{\max} = C_{\min}(T_{h,\text{in}} - T_{c,\text{in}})$, a ceiling set by the *inlet temperatures*, which is a statement about what an infinitely large exchanger could do — not about losses.

### What the picture does not say

- **It is a rating result, not a design.** "Given this hardware, what happens?" is the question answered. "What hardware do I need?" is a different loop, with pressure drop and geometry in it.
- **No pressure drop, no fouling, no cost.** The chart's flat tail says extra area stops paying in duty; it says nothing about what the area costs, and nothing about the pumping power the extra length consumes.
- **Heat-capacity rates are taken as constant** across the pass. A stream whose $c_p$ swings, or which changes phase part way, is outside the assumption.
- **Three configurations only.** Cross-flow arrangements, and the correction factors that go with them, are not offered here.

### Where the derivation lives

**Derivation.** Theory Guide, *Two-stream heat exchanger: the $\varepsilon$–NTU rating method* (`ch:hx-entu`) — the definitions of $C_r$ and NTU, the counter-current and co-current closed forms, the duty from $Q_{\max}$, and the $UA\,\text{LMTD} = Q$ self-check. The design problem the rating chart does not answer — geometry, pressure drop, the sizing loop — is the section immediately following it, *Heat exchanger DESIGN*.

## 3.5 Heat integration: the composite curves and the pinch

*Declared anchor:* `ch:pinch`.

### The question

*Before a single exchanger is drawn: what is the least hot utility and the least cold utility this set of streams can possibly need?*

### What to vary, what to watch

The tool draws what the engine's pinch pass published and computes nothing itself: the stream table (every segment's $T_{\text{in}}$, $T_{\text{out}}$, $|Q|$ and $CP = |Q|/|\Delta T|$), the worked Linnhoff–Flower cascade row by row, the composite curves (the cold curve slid by $Q_{C,\min}$, with the two utility intervals, the maximum recovery and the $\Delta T_{\min}$ gap at the pinch marked), the grand composite and the candidate table, a feasibility assessment of each hot–cold pair whose $Q_{\max}$ values are limits of single pairs and are not additive. The engine also *classifies* the problem from the whole cascade: an interior pinch, several pinches, or a threshold problem with no hot or no cold utility, in which case there is no pinch and none is drawn. The net $Q_{H,\min} - Q_{C,\min}$ is the same at every $\Delta T_{\min}$ (it is the cold total minus the hot total), which is a check a student can make on paper.

> **Try it now.** Change nothing but $\Delta T_{\min}$ — from 20 K down to 10, then up to 30. Watch the composite curves slide together and apart, the two targets move in opposite directions, and the pinch temperature itself change. Nothing physical has been altered: not a stream, not a flow, not a temperature. Ask the class what, then, has changed.

- $\Delta T_{\min}$ — the decisive knob, and the one that is a *choice*.
- **The supply temperatures** — move one stream's start and watch whether the pinch stays where it was. Sometimes it does not, and the region boundaries in the candidate table move with it.
- **The flows** — a flow scales that stream's $CP$, which changes the slope of its segment of the composite curve. A steep segment and a flat one are the same duty over different temperature spans.
- **The candidate table** — read the temperature-feasibility column and the pinch-rule column separately. A match may be thermodynamically possible and still violate the $CP$ rule at the pinch.

### The misconception the construction exposes

> **Common pitfall. That the pinch is a property of the plant.** It is a property of the stream data *plus a number the engineer chose*. Turning $\Delta T_{\min}$ moves the pinch temperature and both targets while the process stands still. Students routinely quote "the pinch is at 115°C" as though it had been measured; it was assumed into existence, half of it by the choice of driving force, and the value of that choice is an economic trade-off between exchanger area and utility bill that this pass does not make for you.

Two more, and the third is the one that decides coursework marks:

- **That a target is a design.** The table lists *thermodynamically admissible candidates* and never ranks them. The word "optimal" does not appear anywhere in this pass, and a gate in the test suite enforces that. Reaching the target may require more exchangers, and more awkwardly placed ones, than anybody would build; plot plan, control, start-up and safety are not in the temperature table.
- **That heating below the pinch merely costs a little extra.** It costs *exactly its own duty*, and so does cooling above the pinch, and the two are additive:

$$\sum(\text{heating below}) + \sum(\text{cooling above}) \;=\; Q_{H,\text{current}} - Q_{H,\min}.$$

  The violation diagnostics report both sums, and they are computed by different code from the targets — so the identity closing is a check you can do by hand and the tool can fail. A student who has only the slogan "do not transfer heat across the pinch" cannot say how much a violation costs; this identity says exactly how much.

### What the picture does not say

- **Energy targets only.** Area targets, capital targets, and the capital-versus-energy trade-off that would let you *choose* $\Delta T_{\min}$ economically are a further phase and are not implemented. The Theory Guide states this in the same words rather than describing something the engine does not have.
- **It never rewrites the network.** The flowsheet you wrote is the flowsheet that ran; the pass analyses it and stops.
- **A unit enters as its net duty.** A column with a reboiler and a condenser is a single net number unless you model the two as separate units — and for a column that is usually the wrong picture. Splitting is the fix, and it is your decision, not the pass's.
- **The grand composite curve is derived in view.** The engine publishes the hot, cold and shifted composite curves; the tool builds the grand composite from those, and labels it as such. The cross-check chip exists so you can see that the derivation reproduces the engine's targets.

### Where the derivation lives

**Derivation.** Theory Guide, *Pinch analysis: the target before the network* (`ch:pinch`) — the $CP$ line, why $\Delta T_{\min}$ shifts the axes, the problem table cascade by cascade, the composite curves, the candidate matches, the violation identity, and an explicit *What this pass does not do.*

## 3.6 Reactor sizing: the Levenspiel plot

*Declared anchor:* `ch:pfr`.

### The question

*The same reaction, the same feed, the same conversion — which reactor is smaller, a plug-flow tube or a stirred tank, and by how much?*

### What to vary, what to watch

One chart carries $1/(-r_A)$ against conversion $X_A$, and on it two areas: the plug-flow reactor's integral $\int_0^{X_{\text{out}}} \mathrm{d}X/(-r_A)$ under the curve, and the stirred tank's rectangle of height $1/(-r_A)$ *at the outlet* and width $X_{\text{out}}$. Both witnesses are twins — byte-identical feed, kinetics and thermodynamics — so the comparison is between reactors and never between chemistries.

> **Try it now.** Raise the plug-flow reactor's volume so $X_{\text{out}}$ climbs, and watch the rectangle and the area separate. At low conversion they nearly coincide; at high conversion the rectangle towers over the area. Then lower the temperature: the whole curve lifts (a slower rate is a larger ordinate) and both reactors grow — but the *ratio* between them is governed by the curve's shape, not its height.

- **Reactor temperature** — through the Arrhenius constant.
- **The two feed flows** — the limiting reactant sets $F_{A0}$ and the co-reactant changes what the rate law sees.
- **The plug-flow volume** — the knob that walks $X_{\text{out}}$ along the curve, which is the axis the whole lesson lives on.

### The misconception the construction exposes

> **Common pitfall. That the stirred tank needs more volume because it is somehow a worse reactor.** Students reach for a hardware explanation — poorer mixing, dead zones, short-circuiting. The chart shows that the cause is the opposite and is purely geometric: a stirred tank is *perfectly* mixed, so every molecule in it sits at the outlet composition and therefore at the outlet rate — the *lowest* rate anywhere in the duty. The tube sees the whole range of rates from inlet to outlet. The rectangle circumscribes the area because perfect mixing throws away every fast rate the reaction was willing to give. *Being well mixed is what costs the volume.*

And immediately after that, the guard against over-learning it:

- **That "a CSTR always needs more volume" is a law.** It follows from the curve *rising* with conversion, which is what positive-order kinetics do. Draw a rate law whose $1/(-r_A)$ has a minimum — autocatalysis, strong product inhibition — and the rectangle can be the smaller of the two over part of the range, which is why real plants sometimes put a tank first and a tube after it. The construction is valuable exactly because it is a picture and not a rule: it shows you how to check, on any kinetics, rather than what to remember.
- **That the smaller reactor is the cheaper one.** Volume is not price. A large stirred tank can be considerably cheaper than a small tubular reactor with the same duty once you count fabrication, catalyst containment, cleaning and control.

### What the picture does not say

- **The comparison is isothermal.** Both witnesses run at the same declared temperature. Nothing here shows the thermal behaviour that makes the two reactors genuinely different animals — for that, go to §3.7.
- **No residence-time distribution.** Plug flow and perfect mixing are the two ideal limits; every real reactor is between them, and neither line on this chart is a real vessel.
- **Reversible kinetics are clipped, and the clipping is announced.** Past equilibrium the engine's rate goes negative and those points are excluded from the area, with the exclusion stated in the chart annotation. An integral run through them would be arithmetic, not physics.
- **The stirred tank's volume is not a knob here**, for a recorded reason: the witness declares it inline, out of reach of the override grammar the classroom knobs use. It always runs its authored value. This is stated in the tool's own provenance line rather than left to be discovered.

### Where the derivation lives

**Derivation.** Theory Guide, *Plug-flow reactor* (`ch:pfr`) — the differential balance, the RK4 integration, and the first-order analytic check that gives the canonical comparison at $k\tau = 1$ ($X^{\text{PFR}} = 1 - e^{-1} = 0.632$ against $X^{\text{CSTR}} = 1/(1 + k\tau) = 0.500$), which is this construction in closed form for one special case. The stirred tank's own extent formulation is in *Continuous stirred-tank reactor* (`ch:cstr`).

## 3.7 Multiple steady states: the Van Heerden diagram

*Declared anchor:* `ch:cstr`.

## The question

*The same reactor, the same feed, the same everything — so why does it sometimes sit cold and sometimes run hot, and could it be made to sit in between?*

## What to vary, what to watch

The diagram is heat **generated** $G(T)$ against heat **removed** $R(T)$, both drawn from the engine's own scan of the reactor's steady-state equation. Generation is a sigmoid: the rate rises exponentially with temperature and then saturates once the reactant is gone. Removal is a straight line, because in an adiabatic vessel the only sink is the sensible heat that leaves with the product. A straight line can cut a sigmoid three times.

> **Try it now.** Drag `T_guess` slowly from 280 K to 430 K and watch the *reported* marker. For a long stretch nothing happens; then it jumps to another crossing on a diagram that has not moved at all. Same equipment, same feed, same equations — a different answer, decided by where you started. That is not a numerical artefact; it is why real reactors have a start-up procedure.

- `T_guess` — the decisive knob. The engine finds *all* the roots, prints them, and reports the one nearest your guess.
- **Feed temperature** — slides the removal line. Push it up far enough and the two lower crossings merge and vanish; the reactor can then only be ignited.
- **Reactor volume** — stretches the generation sigmoid against a removal line that does not move with it.
- **The `steadyStates` count and the ignition span** — how many crossings there are, and how far the reactor jumps between the coldest and the hottest.

## The misconception the construction exposes

> **Common pitfall. That three answers means the solver is broken.** Students trained on "the equation has *an* answer" read multiplicity as a defect — a bad initial guess, a tolerance too loose, a bug to be reported. It is the physics of the system, and the diagram is the proof: an exponential generation curve and a linear removal line intersect three times over a range of conditions, and each intersection is a genuine steady state of the same hardware. The initial guess is not a numerical convenience in this problem. It is the start-up procedure, written as a number.

The one that is harder, and better:

- **That the middle state is merely difficult to converge to.** It is not difficult; it is *unstable*. A steady-state solver will sit on it happily, which is exactly what makes it dangerous pedagogy: the number exists and the reactor cannot hold it. Nudge the temperature up and generation exceeds removal, so it climbs further; nudge it down and it falls to the extinguished state. "The solver converged" and "the plant can run there" are different claims, and this is the cleanest demonstration in the corpus that the second does not follow from the first.
- **That the ignited state is the one you want.** It is the one with the conversion. It may also be past a selectivity limit, past a catalyst's sintering temperature, or past the vessel's design pressure — none of which appear on a diagram whose only axes are temperature and heat.

## What the picture does not say

- **It is a steady-state diagram.** It says which states *exist*. It does not say how the reactor travels between them, how long ignition takes, whether it overshoots, or what the transient does to the catalyst. That is the dynamic simulator's subject.
- **The instability flag is an ordering, not an analysis.** With exactly three roots the tool marks the middle one, which is arithmetic on the engine's own root list. With any other count it makes *no* stability claim at all — a restraint worth pointing out to a class, because it is the difference between a result and a habit.
- **The removal line is straight because this reactor is adiabatic.** Add a cooling jacket at a fixed coolant temperature and it is still straight, with a different slope and intercept; let the coolant temperature vary along the vessel and it is not straight at all, and the geometry of "how many crossings" has to be re-argued.
- **The kinetics are a declared teaching variant.** The witness's own header says so, and its Arrhenius parameters are deliberately *not* knobs: a construction about start-up should not silently become a construction about a different chemistry.

### Where the derivation lives

**Derivation.** Theory Guide, *Continuous stirred-tank reactor* (`ch:cstr`) — the material balance in the reaction extent and the one-dimensional solve the engine performs, which is the equation whose roots this diagram displays. The witness case `tutorials/steady/reactors/cstr05_multiplicity` carries the generation-versus-removal argument in its own `flowsheetDict` header, and is the fullest written statement of the construction in the tree; the tool quotes it. (That witness no longer exhibits three roots: after the acetic-acid correction its liquid esterification is nearly thermoneutral and the scan finds one; a replacement exothermic witness is planned.)

# 4 The correlation library, and how to read a disagreement

Every construction in §3 rests on correlations, and a simulator normally hides them. It picks one, computes with it, and prints a number. The student learns that the friction factor *is* 0.0223, in the same tone of voice the simulator uses for the molar mass of water.

It is not. It is one correlation's answer, and which one was chosen is part of the answer.

This section is about the instruments that make that visible. They are not charts — they run under `choupoProps` and print — but they teach something no chart can, because the thing to be seen is a *disagreement between authorities*, and a chart of one curve cannot show it.

## 4.1 What a correlation carries

A correlation in Choupo is an object, and it is required to answer four questions about itself:

- **What am I?** — the expression, evaluated.
- **Where do I stop being valid?** — the window its own authors stated, in words, printed on every run.
- **Who says so?** — a dated primary citation. A correlation whose source lives in a code comment is one the reader cannot check.
- **Can I still prove it?** — a *verify* against a published anchor, re-run every time, with the deviation printed beside the source.

The fourth is the unusual one. A self-check that nobody wires up is a promise, not a check; running it on every use is what turns it into one.

## 4.2 The friction factor: four answers to one question

*Declared anchor:* `ch:hydraulics-pressure`.

### The question

*Water at* $\mathrm{Re} = 10^5$ *in a commercial steel line,* $\varepsilon/D = 10^{-3}$. *What is the Darcy friction factor?*

Run `tutorials/props/hydraulics/moody01_friction_correlations` and four correlations answer:

| correlation | $f$ | |
|---|---|---|
| Blasius | 0.017770 | **outside its window** |
| Churchill | 0.022343 | |
| Colebrook | 0.022175 | |
| Haaland | 0.021966 | |
| spread over all four | | 25.74 % |
| spread over the three inside their windows | | 1.72 % |

### The two numbers, and why there are two

Read the last two rows together, because either one alone is misleading.

*The correlations disagree by 26 %* is what the first row says, and it is false. Three of them agree within 1.7 % — which for a correlated quantity is agreement, not disagreement.

What produces the 25.74 % is Blasius, and Blasius is not wrong. It is being *misused*. Blasius carries no roughness term at all: it was fitted to smooth pipes and knows nothing about a wall. Asked about $\varepsilon/D = 10^{-3}$ it answers anyway, confidently, twenty per cent low, and there is nothing about the number 0.017770 that looks wrong.

That is the whole content of a validity window, and it is why this run prints one for every correlation on every call. The window is not bureaucracy attached to the formula. It is the only thing standing between a plausible number and a reader.

> **Try it now.** Set `relRough` to 0 — a hydraulically smooth pipe — and run again. Blasius comes back inside its window, the two spreads collapse onto each other, and the four answers agree to a few per cent. Now raise `relRough` to $10^{-2}$ and watch the gap between the two spreads widen: the more the wall matters, the more expensive Blasius's silence about it becomes.
> Then set `Re` to 1500. Three of the four announce that they have substituted the exact laminar law $f = 64/\mathrm{Re}$,

because they are turbulent fits and say so. Churchill does not need to: its single expression contains the laminar law as its own limit, which is why it is the default and why its self-check is anchored there.

### What the anchors actually test — and it differs

The run prints a deviation for each correlation against a published anchor. It would be easy to read four small numbers as four validations. They are not the same kind of statement, and the output says which is which:

- **Blasius** is checked against its own closed form. This tests the *arithmetic* and nothing else — and the run says so, so that a deviation of 0.000 % cannot be mistaken for agreement with a measurement.
- **Colebrook** is checked in the fully-rough limit, where its implicit equation collapses to a closed form. This tests that the fixed-point iteration *arrives*.
- **Haaland** is checked against *Colebrook*, because agreement within about 2 % is precisely the claim Haaland published. Anchoring him to anything else would be checking a claim nobody made.
- **Churchill** is checked against the exact laminar law. This is the strongest of the four: it is a property the expression was built to have and that no coding error preserves by accident.

### What the picture does not say

**None of these four is checked against experimental data.** Choupo carries no friction measurements at all. Two of the anchors are self-consistency by construction and one is agreement with another correlation; only Colebrook's tests an algorithm against a closed form the equation itself provides. The run passes, and what it establishes is that the code computes what the papers wrote — not that the papers are right.

The bench also never says which correlation to use. That depends on the pipe: its roughness, whether the flow is developed, whether the duct is even circular. A tool that ranked them would be making the engineer's decision quietly, which is exactly what a silent default does.

And nothing here speaks about the transition band, $2300 < \mathrm{Re} < 4000$, where no correlation was regressed and the flow itself is not single-valued. Correlations evaluated there are marked outside their window, and that mark is the entire answer available.

## 4.3 One family so far

This discipline covers the convective heat-transfer correlations (`heatTransferBench`) and, since 2026-08-25, the friction factor (`frictionBench`). It does *not* yet cover the rest: the engine computes Wilke–Chang, Chilton–Colburn, Thiele, Weisz, Joback, Lee–Kesler, Rackett and Watson, and those remain internal to the operations that use them — correct, cited in comments, and invisible to a student.

Saying so is the point. A guide that implied an exhaustive catalogue where two families exist would be committing the error this whole section is about: presenting one answer as though it were the only one.

# 5 One number for “like dissolves like”

A student asked to pick a solvent has a shelf of bottles and no way to rank them. The oldest useful answer is a single number per liquid, and it is worth teaching precisely because it is *almost* right — the places where it fails are as instructive as the places where it works.

## 5.1 Where the number comes from

Vaporising a liquid pulls its molecules apart. The energy that costs, per unit volume, is the *cohesive energy density*; its square root is Hildebrand’s solubility parameter:

$$\delta \;=\; \sqrt{\frac{\Delta H_{\mathrm{vap}}(T) - RT}{V_m}} \tag{1}$$

Two things in that expression are worth stopping on.

The $RT$ is the work the vapour does pushing the atmosphere aside. Subtracting it turns an *enthalpy* of vaporisation into an *energy* of vaporisation, and only the second one measures cohesion. Leaving it in gives a $\delta$ a few per cent high — small enough to look right, which is what makes it a favourite error.

And $\delta$ is a function of temperature that nobody writes the temperature next to. As a liquid approaches its critical point the cohesive energy goes to zero, and so does $\delta$: there is no longer a liquid to pull apart. Published tables are at 25 °C; a $\delta$ quoted without a temperature is a number from an unstated state.

## 5.2 Running it

`choupoProps` with a `solubilityParameter` operation.

Choupo *derives* $\delta$ — it is never stored. Every input is already in the component record (the latent heat at $T_b$, the critical temperature, the liquid molar volume), so a stored $\delta$ would be a second home for a fact the record already fixes, free to drift from it. The latent heat at $T$ comes from the same Watson correlation the enthalpy legs use, not a second copy of it.

The output has three parts: what each substance’s $\delta$ is and what it was computed from; how far that sits from an independently published value where the record carries one; and the pairwise table $|\delta_i - \delta_j|$, which is what the question actually was.

## 5.3 The column that makes it checkable

**Try it now.** Look at the *published* column before you look at anything else.
Equation 1 has three chances to go quietly wrong: a unit slip in $V_m$, the missing $RT$, and reporting $\mathrm{Pa}^{0.5}$ where the literature uses $\mathrm{MPa}^{0.5}$ — a factor of 31.6 that still looks like a solubility parameter. None of the three raises an error.
What catches them is that some records carry a $\delta$ somebody else computed, independently, and the run prints the deviation. Across the substances where both exist the derivation reproduces the published value to well under a per cent; a deviation of several per cent is therefore a *finding* and not noise.
This is the same device as the $Z_c$ column in a property data bank: a redundant determination turns a plausible wrong number into a visible disagreement. A tool that computes something you cannot check is a tool that is right until the day it is not.

## 5.4 Where one number stops

**Hildebrand cannot separate the hydrogen bonding it contains.** The cohesive energy it measures *includes* the hydrogen bonds broken on boiling; what one number cannot do is tell that part from dispersion and polarity. Ethanol ($\delta \approx 26$ $\mathrm{MPa}^{1/2}$) and nitromethane ($\approx 25.1$) sit almost on top of each other, and a formulator knows they behave differently — one donates hydrogen bonds and the other does not. The theory cannot express the difference, because it compressed every kind of intermolecular force into one scalar.

That is why the coatings industry uses Hansen’s three-way split: dispersion, polar and hydrogen-bonding contributions, with miscibility judged by a distance in that space rather than on a line. Choupo reads the split from a `hansen {}` block on a record and measures that distance with the `hansenMixture` operation — the EduTool *Choosing a solvent, or a solvent mixture* draws it; the catalogue carries no split itself, and the witnesses carry theirs case-locally with a citation per value.

So the honest reading of a close pair is *candidate*, never *verdict* — and the run prints exactly that sentence, on every call, rather than letting a small number imply more than it carries.

**Try it now.** Put water in the list. Its $\delta$ is enormous ($\approx 48$ MPa$^{1/2}$), far from every organic on the shelf, and the pairwise table will duly report that water mixes with nothing. Water and ethanol are of course miscible in all proportions.
That single row is the clearest lesson the tool gives: the number is a measure of *how much* energy holds a liquid together, and says nothing about *what kind*.

## 5.5 Adsorption isotherms from laboratory data

*Declared anchor:* `sec:adsorption-isotherm-fitting`.

*Tool:* `tool=adsorption-isotherms`. *Case it starts from:* none — the page fits points it generates or points you paste, and runs no engine.

### The question

*I have laboratory equilibrium points $(c, q)$ for a solute on an adsorbent; which isotherm describes them, with what parameters, and how far can I trust it?*

The page opens on what an isotherm is — the loading on the solid against the concentration left in the liquid, at one temperature, each point read from a flask balance $q = V(c_0 - c)/M$ — and on the three shapes (favourable, linear, unfavourable) and what each does to a front in a bed, which is where the `breakthrough` page continues. It then derives the three models: linear $q = Kc$; Langmuir $q = q_{\max}c/(K_L + c)$ from a site balance, with its equivalent affinity form $b = 1/K_L$ and its Henry limit $q_{\max}/K_L$; and Freundlich $q = K_F c^n$ (the exponent written as the course text writes it; Seader writes it $1/n$), which has neither a Henry limit nor a plateau.

### What the page computes, and what it does not

Everything is the page's own arithmetic, marked *page* on every panel. The engine carries Henry and Langmuir isotherms for a gas, in partial pressure, and fits them to measured $(T, p, q)$ data; it has no liquid-phase isotherm and no Freundlich model, and handing it a concentration as a pressure would be a units lie. The page fits each model two ways: the classical straight lines ($1/q$ against $1/c$; $\log q$ against $\log c$) and a direct least-squares fit on $q$ by a hand-written Levenberg–Marquardt, with absolute or relative residuals. It prints each parameter with its standard error from $s^2(J^{\mathsf{T}}WJ)^{-1}$, the root-mean-square residual, the largest relative error, a residual plot, and a diagnostic of whether the measured range can tell the models apart: the decades of $c$ spanned, the fraction of the fitted Langmuir plateau reached at the highest and the lowest point, and how many standard errors the Freundlich exponent lies from 1. Every plot shades the concentrations outside the measured range. Finally the fitted isotherm is used once: the single-stage batch balance $Vc_0 + Mq_0 = Vc + Mq$ drawn as an operating line, giving the dose for a target removal, with a warning when the target lies outside the data. The design record `docs/design/adsorption-isotherms-from-laboratory-data.md` holds the measurements and a proposal for the engine side.

### The examples are synthetic

Three example sets are *generated* by the page, not measured, and are labelled so: each states its model, its parameters, its concentrations and its noise (multiplicative, $\exp(\sigma e)$ with a seeded standard normal $e$), so the fit can be checked against the truth. An organic solute on activated carbon, Freundlich-shaped over four decades of $c$; a metal ion on a chelating resin, strongly favourable Langmuir with a clear plateau; and a trace solute generated from a Langmuir isotherm but measured only far below $K_L$, where nothing in the data can tell the three models apart. Your own points are pasted as two columns, in a unit chosen from a short list.

**Try it now.** On the activated-carbon set, compare the Langmuir straight-line fit with the direct one: the double-reciprocal line gives a capacity about four times smaller (13.5 against 58.5 mg/g), because the lowest concentrations dominate it, and neither Langmuir fit describes a set that has no plateau — the Freundlich RMS is about a fifth of the direct Langmuir's. On the trace-solute set the double-reciprocal line has $R^2 = 0.999$ and still returns $q_{\max} = 124$ mg/g where the direct fit gives $64 \pm 29$: a high $R^2$ on the transformed variables says the points lie near a line, not that the capacity is known. Read that standard error and the diagnostic sentence, then switch the residuals from absolute to relative and watch the Freundlich exponent move towards or away from 1.

**Common pitfall. Reading the smallest RMS as the right model.** A two-parameter model always fits at least as well as a one-parameter one. The residual plot — a run of one sign then the other — and the diagnostic of the measured range say more than the RMS alone.

### What this tool does not carry

One temperature: no heat of adsorption, no van’t Hoff extrapolation. One solute: no competition. Equilibrium only: no uptake rate. One batch stage: no multistage dosing and no column. The standard errors rest on the chosen error model and on local linearity.

## 5.6 Batch adsorption in a stirred tank

*Declared anchor:* `sec:batch-adsorption-stirred-tank`.

*Tool:* `tool=batch-adsorption`. *Case it starts from:* none — the page works on an isotherm you choose and runs no engine.

### The question

*I have a tank of solution and an adsorbent with a known isotherm; how much adsorbent do I need, is it better split into stages, and how long does it take?*

The isotherm is one of three example systems with the page’s own illustrative numbers (a Langmuir dye on powdered carbon, a Freundlich aromatic on granular carbon, a linear trace solute), or any linear, Langmuir or Freundlich isotherm you type — for instance the one fitted on the `adsorption-isotherms` page. One basis throughout: $c$ in mg/L, $q$ in mg/g, $V$ in L, $M$ in g.

### What the page computes, and what it does not

Everything is the page’s own arithmetic, marked *page*. The engine’s batch adsorber and fixed bed contact a gas, in partial pressure, with Henry or Langmuir isotherms; it has no liquid-phase adsorption and no Freundlich model. The isotherm functions are the `adsorption-isotherms` page’s, imported, not copied. Step by step: the equilibrium of one stage as the crossing of the operating line (slope $-V/M$) with the isotherm, found by bisection on a bracket that always holds and printed beside the closed form where one exists (linear, Langmuir); the dose for a target removal, drawn against the removal on a logarithmic axis, with a pre-loaded adsorbent’s equilibrium floor refused by name; one to four stages in crossflow and in countercurrent, each with its mass balance recomputed, and the best split of a fixed total between two crossflow stages; the uptake in time with the whole resistance in a liquid film, $k_L$ from $Sh = 2$ (the still-liquid bound, which a stirred tank exceeds by one to two orders of magnitude, so the times printed by default are an upper bound; the student may replace it), $a$ from the particle density, in closed form for a linear isotherm and integrated numerically (fourth-order Runge–Kutta) otherwise, with the time to a target at five particle sizes; and the Biot number with Glueckauf’s series resistance, with a warning when diffusion inside the particle makes the film-only times too short. The design record `docs/design/batch-adsorption-in-a-stirred-tank.md` holds the measurements and a proposal for the engine side.

**Try it now.** On the dye example with 50 g in 100 L, read the best two-stage split: about 69 % of the carbon in the first stage, leaving 0.92 mg/L against 1.15 mg/L for half and half. Switch to the linear example and watch the optimum return to exactly one half. Then, in the kinetic step, halve the particle diameter and check that every time falls by four — the still-liquid result, which holds because $Sh$ is fixed — and then declare $Sh = 100$ to see the times a stirred tank gives.

**Common pitfall. Reading $Sh = 2$ as a stirred tank.** It is a sphere in still liquid, and a stirred tank’s Sherwood number is tens to a few hundred for particles of this size (Treybal’s Eq. 11.29 gives about 190 for 0.5 mm at 1 kW/m$^3$; Seader’s Example 15.10 has $k_L = 5 \times 10^{-5}$ m/s), so the default times are one to two orders of magnitude too long. And the Biot number does not fall as the particle shrinks *while Sh is fixed*: at $Sh = 2$ it is $D_{AB}/D_e$, larger than one for every porous particle; in a tank $Sh$ grows with the diameter and so does $Bi$.

### What this tool does not carry

One temperature and no heat of adsorption; one solute; a perfectly mixed tank; no diffusion solved inside the particle (only flagged); no continuous column, which is the `breakthrough` page.

## 5.7 Adsorption: sizing a bed from a breakthrough curve (LUB)

*Declared anchor:* `ch:adsorption`.

*Tool:* `tool=lub-scaleup`. *Witness:* `tutorials/batch/adsorber/batch13_breakthrough_co2`, the same laboratory column the `breakthrough` tool integrates.

## The question

*I have a breakthrough curve from a short laboratory column: outlet over feed concentration against time, at a known length, velocity, feed and packing. The plant needs a bed that holds a given feed flow for a given service time before the outlet goes off specification. How long, what diameter, and how much adsorbent?*

## What to vary, what to watch

The construction is the length-of-unused-bed method, performed on the screen in the order a student performs it on paper, with the engine's own numbers printed beside each step. Two times are read off the curve: the breakthrough time $t_b$ at the declared criterion, and the stoichiometric time $t_{st}$ as the trapezoid area above the curve. The unused length $\mathrm{LUB} = L_{\mathrm{lab}}(1 - t_b/t_{st})$ is carried to full scale unchanged, the equilibrium section is the laboratory column scaled by $t_{\mathrm{req}}/t_{st}$, and the diameter follows from the design flow at the *same* superficial velocity. Two families of knobs sit beside the plot, and the page says which is which: the laboratory column's velocity, length, temperature, pressure and horizon re-run the engine in the browser; the criterion, the service time and the design flow are post-processing and re-draw only, because none of them changes the curve the column produced.

> **Try it now.** Start from the witness. Read the first table: the trapezoid $t_{st}$ and the engine's own analytic $t_{st}$ (its pre-run claim $(L/u)\,R_f$) agree to about seven digits, and the tool says the gap is the mesh plus the dropped tail. Now shorten the horizon knob to 3200 s: the outlet has not reached the feed value, the page flags the data as *incomplete*, and every length in the last table turns into a lower bound. Put the horizon back. Move the criterion from 0.05 to 0.02 and watch $t_b$ slide left while $t_{st}$ does not move: the unused length grows, because a stricter specification switches earlier. Finally raise $t_{\mathrm{req}}$ from 10 000 s to 40 000 s and watch $\eta_{\mathrm{bed}}$ climb toward one — the same few centimetres of unused bed, added once to a bed four times longer.

- **Breakthrough criterion** — a specification, not a property of the bed. It moves $t_b$ and therefore LUB; it never moves $t_{st}$.
- **Required service time** $t_{\mathrm{req}}$ — the knob that sets the equilibrium length, linearly. Watch the utilisation $\eta_{\mathrm{bed}} = L_{es}/L_{\mathrm{full}}$ rise with it.
- **Design feed flow** $Q$ — sets the diameter alone, $D = \sqrt{4Q/(\pi u)}$, and through it the adsorbent mass.
- **Bed length** $L$ (engine knob) — both times move, the band between them does not: LUB is a property of the conditions, and the plot makes that visible.
- **Velocity** $u$ (engine knob) — the one knob that changes LUB itself, which is exactly why the scale-up holds $u$ fixed.

## The misconception the construction exposes

> **Common pitfall. That the plant bed is the laboratory column scaled by length.** A student who has measured a column that wastes five per cent of itself expects a plant bed to waste five per cent too, and sizes it as $L_{\mathrm{lab}}\,t_{\mathrm{req}}/t_{st}$ with a five per cent margin. The unused length is not a fraction; it is a *length*, set by the width of the mass-transfer zone, and under constant pattern it is the same length in every bed run at these conditions. It is added once. A half-metre column that throws away 2.5 cm looks poor; a three-metre bed with the same 2.5 cm unused wastes under one per cent. Laboratory columns look terrible and plant beds do not, and the page's last table is that sentence as numbers.

A second confusion, the one that costs a plant:

- **That a curve stopped early is a curve.** The stoichiometric time is an *area*, and an area cut before the outlet reached the feed value is too small — so the equilibrium capacity implied by the curve falls short of the isotherm's, and every length sized from it is short. The capacity check in the third step exists for this: the curve's own balance against the engine's retention factor, and a disagreement points at the data, never at the isotherm.

## What the picture does not say

- **Constant pattern is assumed.** A favourable isotherm and a column longer than the zone buy it; a linear or unfavourable isotherm, or a column the zone never fits in, gives a curve the method sizes from wrongly with no warning on the page.
- **Same velocity, temperature and particle size.** LUB is transferred, never scaled. A plant bed at other conditions needs a curve measured at those conditions.
- **Isothermal, single adsorbate, one loading step.** The witness is declared isothermal, feeds one adsorbing species, and starts clean; regeneration and the cycle are other cases' subject (`batch21_tsa_hot_purge`, `batch23_tsa_cycles`), named on the page and not taught by it.

- **The engine's curve is a model.** By default the plot is the engine's integration of the witness on its default second-order van Leer scheme, whose mesh adds about 5 % to the front's width at 100 cells (first-order upwind, which a case may declare, adds about 43 % and makes the unused length 2.5 cm instead of 1.75 cm). A measurement enters through the open case: a `constant/experimental/breakthrough.csv` with a `t_s,c_over_c0` header is drawn as markers beside the engine's curve and put through the same arithmetic, labelled measured. No tutorial ships one today, and the tool says so rather than inventing a curve.

### Where the derivation lives

**Derivation.** Theory Guide, *Adsorption: isotherms, LDF uptake, and the fixed bed* (`ch:adsorption`): the two equations the bed integrates, the retention factor $R_f = \varepsilon + \rho_b q^*(c_{in})/c_{in}$ and the stoichiometric time $t_{st} = (L/u)\,R_f$ the unit announces before it runs, and the mass-transfer zone whose width the unused length measures. The trapezoid, the crossing and the scale-up on this page are the hand method itself, labelled as such, and judged against those engine numbers.

## 5.8 Chromatography: peaks, plates and resolution

*Declared anchor:* `sec:chromatography`.

*Tool:* `tool=chromatography`. *Case it starts from:* none in the corpus — the page writes a `choupoBatch` case of its own (one fixed-bed adsorber unit, a synthetic linear sorbent, a recipe that switches the feed back to carrier) and runs it in the browser.

### The question

*I want to separate two solutes on a column: when does each come out, how wide are the peaks, are they resolved, and what changes when I load more?*

The bed of the breakthrough page is fed a short injection of two solutes instead of a step. The page derives the retention time $t_R = t_0(1 + k')$ with $k' = \rho_b K_c/\varepsilon$, draws the plate model (the Erlang peak of $N$ stirred plates) beside the Gaussian of the same mean and variance, builds the van Deemter curve $H = A + B/v + Cv$ from the bed's own parameters (particle diameter, molecular diffusivity, the linear-driving-force constant), computes the resolution $R_s = (t_{R,2} - t_{R,1})/(2(\sigma_1 + \sigma_2))$ beside the fundamental resolution equation, cuts the outlet in time for purity and recovery, and then loads the column until the isotherm bends.

### What the engine says, and what the page computes

The engine runs the linear pulse: its outlet histories and its own stoichiometric times, Peclet number and mass closure are read off the run, and a run that does not belong to the current knobs is not drawn. The page puts the moments derived in the Theory Guide (elution chromatography, in the adsorption chapter) beside it, with the dispersion the first-order grid adds (one plate per cell) as a column of its own — the page's case declares `convectionScheme upwind;` for exactly that reason, since the engine's default van Leer scheme adds no such term — and its own second-order integration of the same bed equations as a third reading. The overloaded column — a competitive Langmuir isotherm, in concentration units — is the page's integration only, on 100 cells, with its mass ledger printed and a button that repeats it on 200 cells and prints the change. The design record `docs/design/chromatography-peaks-and-resolution.md` holds the moment derivation, the native measurement that decided to use the engine for the linear part, the tests, and a table of every published form the page uses with its checking status.

### What to vary, what to watch

**Try it now.** At the defaults the two peaks overlap ($R_s$ just under 0.9). Raise the bed length to 1.5 m: $R_s$ passes 1.5, but only as $\sqrt{L}$, and the engine's dotted peaks stay wider than the page's Gaussians — that is its grid, one plate per cell, and the extra row in the table accounts for it. On the van Deemter plot move the velocity towards $v_{\mathrm{opt}}$ and watch the plate count rise. Set the plate count of the plate-model panel to 3 and to 100 and compare the gap to the Gaussian. In step 5 raise the load from 0.02 to 1.6: the peaks lean forward and come out earlier; the two solutes are injected in equal amounts, so each lowers the other's loading and both move forward (the first is displaced by the second, and the second is pulled along by the first); switch competition off to see how much of that is competition. The tag-along plateau needs a feed dominated by the first solute, which this page does not inject. The table of step 6 shows the amount recovered at 99 % purity rising with the load and then falling.

**Common pitfall. That a longer column is the way to resolve two peaks.** Resolution grows only as the square root of the plate count, so twice the resolution costs four times the column; the selectivity term $(\alpha - 1)/\alpha$ is usually the cheaper lever, and at high load no column length restores the purity the overload took.

### What this tool does not carry

Liquid chromatography has no engine route: the engine's fixed bed is a gas unit, and the overloaded column is the page's integration of equations in the form a liquid column takes. The velocity is held constant (a poor assumption for a concentrated gas pulse), the band is isothermal, the competition is the competitive Langmuir model with one saturation capacity, the dispersion factors are $\gamma = 0.7$ and $\lambda = 0.5$ (the latter on the low side of the published values, so the page's peaks are somewhat narrower than a real column's), and gradient elution, extra-column broadening and the simulated moving bed are not covered.

## 5.9 Fluidised beds: minimum fluidisation, regimes and bubbles

*Declared anchor:* `sec:fluidised-bed`.

*Tool:* `tool=fluidised-beds`. *Case it starts from:* none — Choupo's engine has no fluidised-bed model, and every number on the page is the page's own arithmetic.

### The question

*I have a bed of particles and a gas: at what velocity does it fluidise, what regime will it be in, how much does it expand, and how much gas bypasses the particles?*

The page draws the bed's pressure drop against the gas velocity: Ergun's line through the packed bed rises until it meets the bed's weight per unit area, and the crossing is the minimum fluidisation velocity, found by bisection and set beside Wen and Yu's correlation with the difference printed. It places the powder in Geldart's groups on a straight-line approximation of the published map, marks the velocities that bound the regimes on a velocity axis (minimum bubbling for a group A powder, the terminal velocity of one particle from its drag balance, and the onsets of turbulent fluidisation and of transport), expands the bed with growing bubbles in the two-phase picture and integrates the bubble-to-cloud exchange to show how much gas bypasses the solid, and lists which size classes of a distribution the gas can carry out to a cyclone. A closing exercise asks the reader to refute three planted statements with the page.

### What the engine says, and what the page computes

The engine has no fluidised bed. The page writes two laws in the same form as the engine: Ergun's gradient of the plug-flow reactor's packed bed, and the single-particle terminal velocity of the pneumatic conveyor on Schiller and Naumann's drag coefficient. Everything else — the correlations of Wen and Yu, Abrahamsen and Geldart, Bi and Grace, Mori and Wen, and Kunii and Levenspiel's exchange coefficient — is evaluated by the page, and every one of their constants is listed in `docs/design/fluidised-bed-hydrodynamics.md` as still to be checked against its source. The gas is declared: its density from the ideal-gas law, its viscosity typed. The design record also proposes what an engine unit for a fluidised bed would need; none is built.

### What to vary, what to watch

**Try it now.** At the defaults (sand-like particles of 200 µm in air) the bed fluidises near 5 cm/s and is a group B powder that bubbles at once. Change the voidage from 0.42 towards 0.415 and watch the gap to Wen and Yu shrink, then to 0.48 and watch it grow: $\varepsilon^3$ sits in both of Ergun's terms. Choose the fine catalyst: a window without bubbles opens between $u_{\mathrm{mf}}$ and $u_{\mathrm{mb}}$. In step 4 raise the velocity and the bed diameter and watch the bypassed share grow as the bubbles grow; compare the row with the bubbles frozen at their size at the distributor.

### The laboratory

A button near the top of the page, *Open the laboratory*, opens a second tab in which the same arithmetic is drawn as one to three columns side by side, one kind of particle in each, under the same gas and one velocity control. Its task, in one sentence: *I turn up the gas on a bed of particles, I watch what the bed does and I read the instruments, and from my own readings I find the velocity at which it fluidises and the one at which it is carried away.* Below the minimum fluidisation velocity the dots sit still at the packed height; above it the bed is drawn at the page's

computed height, with voids of Mori and Wen’s diameter rising at the computed bubble velocity, as many as the computed bubble fraction asks for; above a particle’s terminal velocity the dots leave through the top and the column empties. Each column carries its instruments (pressure drop, bed height, regime), and a notebook records the readings going up and coming down and plots them. The minimum fluidisation velocity and the terminal velocity stay hidden until the reader types the values read off his own points; then the page’s values appear, with the difference. The dots are a drawing, not a simulation: their number, size and motion are not to scale, and how fast a column empties is not computed. The kinds of particle are not mixed in one column, because the page computes each alone and a mixed bed has rules (segregation, a mixture’s minimum fluidisation velocity) it does not have. A cohesive group C powder is drawn packed at every velocity, with the page’s sentence on why. The record is `docs/design/fluidisation-virtual-laboratory.md`.

**Try it now.** In the laboratory, raise the gas slowly and record a reading at each step: the pressure drop of the sand-like column climbs and then stops climbing while its bed starts to swell. Read the knee off your points, type it, and compare with the page. Go on to above the catalyst-like particles’ terminal velocity, watch that column empty, then come back down and record again: the pressure drop no longer returns to where it was, because the bed has gone.

**Common pitfall. That a fluidised bed’s pressure drop grows with the gas flow.** Once the bed is fluidised the gas carries its weight and nothing more; the extra gas goes into bubbles, and the pressure drop across the bed stays where it was.

### What this tool does not carry

Spherical, smooth particles of nearly one size (except in the size table); one gas, isothermal; no wall, no distributor pressure drop, no jets; no elutriation rate and no freeboard; no reactor model. Every correlation is empirical, and the published accuracies are not quoted because they have not been checked.

## 5.10 When the fast obeys the slow: separation of time scales

*Declared anchor:* `ch:stiff-ode`.

*Tool:* `tool=fast-and-slow`. *Cases it starts from:* three pairs, each a full and a reduced model, with one declared number of the full case changed by the slider:

- `unsteady07_fast_tank_full` and `unsteady08_fast_tank_reduced` (in `tutorials/unsteady/`);
- `batch30_fast_equilibrium_full` and `batch31_fast_equilibrium_reduced` (in `tutorials/batch/reactor/`);
- `batch32_fast_intermediate_full` and `batch33_fast_intermediate_reduced` (the same folder).

### The question

*My model is heavy and stiff: which variables can I stop integrating and replace by an algebraic relation, and how wrong will I be?*

The page opens on a puzzle: a plant has more fast motions than anybody could list, yet the formulas engineers design with have two or three variables and fit within a design tolerance. Why are they so simple? The page “Why engineering correlations are simple” names the separation of scales as one mechanism and shows it on a two-variable toy; this page shows it at work on running process models, with the engine computing both sides. The short answer comes first: divide the fast time scale by the slow one, $\varepsilon = \tau_{\text{fast}}/\tau_{\text{slow}}$, replace each fast part by the relation that makes its own rate zero, and the error falls with $\varepsilon$. Three case studies test it, each as a guess, a reveal (the engine runs both models; nothing runs before the student presses the button), a slider that makes the fast part slower until the reduction fails, the cost of the full model, and “what is left when the fast part obeys”: the reduced law, counted against the full one, with the range of the slider where its error stays under a tolerance the student sets. The cases are a small tank behind a big one (the small one replaced by a `storageTank` wrapped as `quasiSteady`), a fast equilibrium feeding a slow step (one pool at $kK/(1+K)$, an `expression` rate law) and a reactive intermediate (a rate law in the stable species alone). The closing section, “So this is why correlations are simple”, puts the three reduced laws beside the effectiveness factor of a pellet, a film coefficient or friction factor, the membrane module solved quasi-steady inside a tank’s start-up, and the engine’s refusal of a loop with no vessel on it, quoted rather than fired.

### What the engine says, and what the page computes

Every curve is the engine’s: `choupoSemiContinuous` for the tanks, `choupoBatch` with the adaptive Rosenbrock23 integrator for the reactors, whose accepted steps the page counts from the step history the engine prints. The page

divides mole numbers by their total, subtracts the reduced curve from the full one, and draws two closed forms as labelled checks: the largest gap of two linear first-order parts, $\varepsilon^{1/(1-\varepsilon)}$, and the stability limit of a fixed explicit RK4 step. The time-integrated driver publishes no step count, so the tank case shows its declared fixed step count and nothing in its place.

### What to vary, what to watch

**Try it now.** At the defaults the largest gaps are 4.3 % (tanks, $\varepsilon = 0.05$), 0.62 % (equilibrium, $k_f/k = 100$) and 0.95 % (intermediate, $k_2/k_1 = 100$). Move each slider to its slow end and watch the gap reach 37 % where the two time scales are equal. In the intermediate case, compare the gap with the tank case at the same $\varepsilon$: the numbers are the same, because the two problems are the same pair of equations. In the reactor cases watch the step count: the stiff integrator's barely moves, while the explicit bound grows with the fast rate. Then set a tolerance and measure the window.

**Common pitfall. That reducing a stiff model is mainly about speed.** With a stiff integrator the full model costs little more; what the reduced law gains is that it no longer needs the fast constants at all, which is why a correlation can be fitted with two or three numbers.

### What this tool does not carry

Linear, first-order models only; synthetic species and author-set numbers; isothermal reactors. Next, named and not built: batch electrodialysis (the film against the emptying tank) and a cascade control loop. The measured tables are in `docs/design/fast-and-slow.md`.

## 5.11 Operator panels and alarms

*Declared anchor:* `sec:operatorPanels`.

*Tool:* `tool=operator-panels`. *Case it starts from:* `batch/electrodialysis/edbatch01_constant_current`, cloned with at most one declared value changed for a disturbance and with the page's own `system/panelDict` in place of the shipped one.

### The question

*My process publishes dozens of variables: what do I put in front of the operator, in what form, and which alarms?*

The batch electrodialysis rig publishes 25 series at 41 written instants. The page shows the same run twice — every number the engine publishes, and the six elements of the panel the case declares — and asks the student what is happening, first from the numbers and then from the panel, under one of four runs: as designed, the rectifier set past the stack's rating, a lower current efficiency, and a richer diluate. It then maps the operator's tasks to the least information that answers each, draws one value four ways (a bare number, on its scale, as a trend, as a lamp), arranges overview, unit and detail, moves an alarm limit, runs a naive panel that puts an alarm on everything that moves, grows a panel in three stages, and ends with an exercise: a panel with five planted defects.

### What the engine says, and what the page computes

Every run, and every fact about a panel, is the engine's: `choupoBatch` reads the panel after the time loop and reports which limit is crossed and when, which element never moves, which limit is never approached, which column is not published, which alarms watch the same column, and how many alarms are active at once. The page writes the panels and draws the worked one as a lesson-local illustration; an element is coloured only where the engine's alarm interval covers the time cursor. The one count the page makes itself — alarm onsets inside any ten minutes, against a handling rate the student sets — is labelled as the page's, on an assumption the page labels. The design record `docs/design/operator-panels-good-practice.md` holds the native runs, the tests and the table of every practice rule with its checking status.

### What to vary, what to watch

**Try it now.** Pick the lower current efficiency and move the cursor to the end: nothing alarms, and the demineralisation trend's ideal line falls with the run, because the unit computes it with the same efficiency — the reference moved with the fault. Pick the rectifier set too high: the current-density bar is in alarm from the first second and the lamp lights at 1150 s instead of 1600 s. In step 5 move the limit to 0.25 and to 3 and read the facts `alwaysViolated` and `limitOutsideRange`; then run the naive panel on the designed run and count the alarms that start together near the end.

**Common pitfall. That more information makes a better panel.** Every element costs the operator's attention, and every alarm interrupts him. A panel that shows everything shows nothing, and a panel where everything alarms has no alarm.

### What this tool does not carry

Nothing on the page is a grade. Only the first slice of the panel exists in the engine: there is no workspace that draws a case's panel, no draft, no tuning and no lock yet. The unit takes no change in the middle of a run, so a disturbance is a second run. The facts are resolved to the 50 s write grid. The practice of alarm management and operator displays is written from reasoning and not checked against the published standards, which were not read; the alarm-handling rate is an assumption of the page.

## 5.12 Membranes: a UF/NF batch, concentrated and washed

*Declared anchor:* `ch:membrane`.

*Tool:* `tool=batch-membrane`. *Witnesses:* `tutorials/batch/membrane/diafilter01_nf_desalting` and its twin `tutorials/batch/membrane/diafilter02_fouling_decline`, which differ by a `fouling` block and by nothing else.

### The question

*I have a stirred vessel behind a membrane. I can take permeate away and let the vessel shrink, or I can make solvent up as fast as it leaves and hold the volume. The first concentrates what the membrane keeps; the second washes out what it does not. How far does each one get, how long does it take, and how much of the product do I lose doing it?*

### What to vary, what to watch

The two modes are one line of the model apart — the diafiltrate rate $Q_d$ is zero in a concentration and equal to the permeate rate in a wash — and the page makes that the mode switch. It also makes it the *axis*: each mode is plotted against its own clock, the volume concentration factor $\mathrm{VCF} = V_0/V$ for a concentration and the diavolumes $N = \int Q_d \,\mathrm{d}t/V_0$ for a wash. Neither clock is seconds, and that is deliberate: both are dimensionless, both start at their own origin, and both make runs at different areas and pressures comparable, which seconds do not. Flipping the mode therefore changes what the horizontal axis *means*, not merely the curve on it.

Four knobs re-run the engine in the browser — the feed pressure, the membrane area, the film coefficient $k_{\mathrm{film}}$ and the horizon — beside two switches, the mode and whether Hermia's cake law is applied. Every retained fraction is drawn on a logarithmic axis, because the closed forms the construction is about are exponentials and a student should be able to read a straight line where one is claimed.

**Try it now.** Start on the wash. The classical result says the retained fraction falls as $\exp\big[-(1-R)N\big]$: a straight line on the log axis, whose slope is fixed by a *constant* rejection. Put the engine's curve beside it and the two part. Now open the rejection trace underneath: $R_{\mathrm{obs}}$ is not constant, it climbs as the salt leaves — and the ideal line the page drew was built from *this run's own initial* $R$, so the gap cannot have been arranged. Switch to concentrate mode and watch the axis become VCF and the law become a power. Watch which power: the RETAINED fraction falls as $\mathrm{VCF}^{-(1-R)}$ while the concentration rises as $\mathrm{VCF}^{R}$, and at $R = 1$ that second form gives $c/c_0 = \mathrm{VCF}$ — a membrane that keeps everything concentrates exactly as the vessel shrinks. The same assumption as the wash, spent the same way. Then turn on Hermia cake and change nothing else: the twin case is the same operation with a deposited layer, and the run buys fewer diavolumes in the same horizon. Finally look at the last plot — the product $J_w c$ against VCF — and find where a wash is cheapest.

- **Mode** — the one line that differs, $Q_d = 0$ or $Q_d = Q_p$. It changes the question, the clock and the axis.

- **Feed pressure** — raises the flux and therefore shortens every diavolume, but it also raises the wall concentration, which is where the rejection's drift comes from.
- **Membrane area** $A$ — buys time and nothing else: the trajectory against $N$ or VCF barely moves, while the clock in seconds does. That is the clocks' whole argument, made visible with one knob.
- **Film coefficient** $k_{\text{film}}$ — the stirring. Lower it and the wall concentrates, the osmotic pressure rises and the flux falls, with no fouling anywhere in the case.
- **Hermia cake** — a *claim about a mechanism*, not a fitted exponent, which is why the engine makes the case state a `reason` in writing before it will apply one.

### The trade, and the one classical optimum on the page

Washing longer buys purity and costs two things. It costs product, because no rejection is one and the exponent never stops running; and it costs time, because every diavolume is a vessel volume of permeate that must cross the membrane, $t_{\text{DF}} = NV/(AJ_w)$. At a fixed inventory of the retained species a smaller vessel is a more concentrated one, so pre-concentrating shortens every diavolume — and simultaneously raises the osmotic pressure and the polarisation, which cuts the flux. The two pull against each other, so a cheapest concentration exists, and the classical statement of it is that it sits where the product of flux and retained concentration is largest: Ng, Lundblad and Mitra, *Optimization of Solute Separation by Diafiltration*, Separation Science **11**(5) (1976) 499–502.

The page applies that *criterion* and deliberately not the closed form usually quoted beside it. “Wash at the gel concentration divided by e” does not follow from the criterion alone — it follows from the gel-polarised flux law of ultrafiltration, and the run on this page does not carry that law: its flux is an applied pressure less an osmotic pressure through the declared solution-diffusion model (the engine has the gel law as `transport gelPolarisation`, witness `tutorials/steady/membranes/membrane19_uf_gel_polarisation`, but this diafiltration does not declare it). So the tool reads the product off the engine's own concentration trajectory, marks the maximum, and says whether that maximum was bracketed by the sweep or sitting on one of its ends, because an endpoint is not an optimum — it is a run that did not find one. Nothing on the page borrows a constant from a model that is not running.

### The misconception the construction exposes

**Common pitfall. That the rejection is a property of the membrane.** It is written as a single number on a datasheet, it is taught as a single number, and every closed form on this page is obtained by carrying it outside an integral. It is not a number: it is an outcome of the wall concentration, which follows the flux, which follows the osmotic pressure, which follows a composition that the run is deliberately changing. A student who sizes a wash from the datasheet's $R$ will over-wash or under-wash, and will not know which. The engine publishes $R_{\text{obs}}$ at every instant for exactly this reason, and the page draws the hand result from the run's own *initial* rejection so that the two curves start together and part for a reason the student can watch. Where they stay together, the hand derivation was safe — and knowing which case you are in is the whole skill.

**Common pitfall. That flux decline means fouling.** Lower the film coefficient in a case with no `fouling` block at all and the flux still falls, because the wall concentrates and the osmotic pressure rises. Concentration polarisation is reversible and instantaneous; fouling is an accumulated history of what the membrane has been asked to do. The two twin cases exist so that the difference can be seen rather than asserted: `diafilter01` and `diafilter02` are the same operation, and one block apart.

### What this tool does not carry

Said here because the page says it too, and an absence that is named is worth more than a green run that implies otherwise. A *feed-and-bleed membrane loop* is not a mode of this unit: it is a steady module inside a recirculating tank, which Choupo builds as a flowsheet — a steady recycle at steady state (`tutorials/steady/membranes/membrane18_nf_feed_and_bleed`) and, in time, the module wrapped as `quasiSteady` beside a tank under `choupoSemiContinuous` (`tutorials/unsteady/unsteady03_nf_`). The tree carries no measured fouling data at all, so a cake constant is equipment-and-feed data a case declares and never a catalogue value. The polarisation coefficient is declared, never correlated from a geometry the engine was not told. The pump work that drives the filtration is not ledgered, so the campaign's first law reports itself UNAVAILABLE rather than closing on a term quietly set to zero. And the vessel is one well-mixed liquid at one temperature.

## 5.13 The tree of chemistry: a hundred products, eleven roots

*Declared anchor:* `ch:balances`.

*Tool:* `tool=tree-of-chemistry`. *Records:* the folder `data/standards/routes/` (its `README.md` is the grammar); no engine run, and no case is needed to read the page.

## The question

*The list of commodity chemicals is a hundred names long and looks like a hundred unrelated industries. Is it? And if the energy system changes, how much of that list has to be rebuilt?*

The answer the page carries is that the list is one tree. The products grow from seven organic roots — ethylene, propylene, the C4 olefins, benzene, toluene, the xylenes, and synthesis gas with methanol as its first derivative — and four inorganic ones — ammonia, sulfuric acid, chlorine with caustic soda, and the nitric and phosphoric acids — and the sustainable transition replaces the *roots*, not the leaves. The graph is a curated record: substances are nodes, a named production route is an edge, and each edge is a balanced equation (or, for a cracker whose feed has no formula, a stated slate) with a yield band and a citation.

## Three views of one graph

- **Depth.** Pick a product. The page walks the routes backwards to the roots and feedstocks and prints, on every edge, the kilograms of each input per kilogram of what that route makes. For a reaction

$$m_i = \frac{|\nu_i|\, M_i}{\nu_t\, M_t\, y},$$

  the floor $|\nu_i| M_i/(\nu_t M_t)$ from the balanced equation and the molar masses of the component records, divided by the route's yield $y$. Because $y$ is a *band*, so is $m_i$, and along a chain the bands multiply. The page starts with three worked products — ethylene and chlorine per kilogram of PVC, the Hock route to phenol from benzene and propylene, and methanol from $CO_2$ with the hydrogen from water — and each can be changed by choosing another route for any substance on the path.
- **Convergence and forced co-production.** Pick a root or any substance. The page lists what consumes it, everything reachable from it, and the products at the leaves nothing else consumes. Pick a route and it lists what the route makes whether anyone wants it or not: every product that is not its `target`. The cumene route's acetone follows phenol at a fixed molar ratio; a steam cracker's propylene, C4 cut and pyrolysis gasoline follow its ethylene in the proportions of a slate.
- **The tree of 2050.** For each root, the incumbent routes beside the transition routes, with hydrogen per carbon derived from the stoichiometry as $-\nu_{H_2}/(\nu_t n_C)$ (three for $CO_2$ to methanol, four for $CO_2$ to methane) and the energy figures where a source states one. The ammonia edge is written once: green ammonia is the same reaction fed by a different hydrogen route, so what the page shows replaced is the *feeder*.

## What to vary, what to watch

**Try it now.** Open PVC and read the two totals. The floor of one ethylene and one chlorine per repeat unit is already about 0.45 and 1.13 kilograms per kilogram; the bands sit above it, and the table says by how much. Now click the dichloroethane box and make it by oxychlorination instead of direct chlorination. The chlorine does not go away: the new step takes its chlorine as HCl, which the page makes from chlorine and hydrogen, so hydrogen and oxygen join the totals. The record marks that route as a declared cycle — the balanced plant feeds it the HCl its own cracker releases — and the page does *not* net that HCl against the chlorine; it states the limit. Then open phenol, go to the forced co-production view and read the acetone beside it: one mole per mole, so about 0.62 kilograms per kilogram, and a single number rather than a band, because the yield does not enter that ratio.

## The numbers are estimates, and the page draws them as such

**Common pitfall. That a figure on the page was measured.** Every yield band and every slate share in the records is the curator's estimate (`origin assumed`, with a reason); none was read back against a published table. The page draws each as an estimate — a dashed underline, the mark “est.”, the reason on hover and again in the detail row of the selected route — and says so on its first screen. The stoichiometry is checked atom by atom by `bin/curate/check_chemistry_graph.py`; the only figures read from a source are the IEA's production volumes and four ammonia energy intensities, each with its page. A route with a Choupo case links to it; a route without one says “no case yet” and links nothing.

### What this tool does not carry

The graph is a selection of the chains a course on the chemical industry names, not an inventory, and no check can see a route that is missing. It holds no prices, no capacities beyond the IEA's five volumes, and no environmental inventory, so it does not say which route wins; the Choupo cases linked from the routes, and the route-screening tool, are where that is asked. Co-products are listed, not credited.

## 5.14 Patent expiry and generic entry

*Declared anchor:* `ch:economics`.

*Tool:* `tool=patent-expiry`. *Case it starts from:* `tutorials/batch/plant/apalutamide01_tablets` (and, with other numbers, `donepezil01_tablets`); no engine run.

### The question

*My group designs a generic tablet plant for the day the originator's exclusivity ends. What volume will the plant sell, at what price, year by year, what does the case's economics read from that, and how much of the revenue is at risk if we enter a year or two late or the price falls further than we assumed?*

In this industry the business case is decided by a calendar, not by the equipment. The page takes the originator's annual sales and its price before exclusivity ends, which fix the volume $V_0 = S_0/P_0$ of the molecule, and then applies four curves that you can read and edit: the generic price relative to the originator's, either by the number of generic producers or by the years since the first generic entry; the cumulative share of the volume the generics take; the number of rivals in each year; and the share of the generic volume the newcomer wins. In market year $k$ the plant sells $N = \min(V_0\, g_k\, w(n_k), Q)$ tablets at $p = P_0\, \rho\, f$, with $g_k$ the generic share averaged over the year (the curve is read at the *end* of a year), $w$ the newcomer's share among $n_k$ producers, $\rho$ the price ratio, $f$ the depth factor of the sensitivity and $Q$ the design production.

### What the case reads

The economics block holds one price per kilogram and one production; the volume profile is its `rampUp` list, the fraction of the design production sold in each of the first operating years (`src/postProcessing/EconomicsPass.cpp` requires each entry in $[0, 1]$ and at most `projectLife` entries). The page prints that list and the one constant price $p^* = \mathrm{PV}/\sum_j N_j(1+i)^{-j}$ that gives the same present value as the falling prices, in the case's own units (EUR per kilogram of `packagedTablets`). The equivalence holds for the discount rate you gave the page; change the rate in the case and re-run the page.

### What to vary, what to watch

**Try it now.** Read the table first. The default curves bring the plant to four producers in year three and a price of about 4.0 EUR a tablet, the case's own price, but the volume reaches only about ten million tablets a year, below the fourteen million the case produces from its first year: the case's 25 % market share and this page's curves do not agree, and the column “sold” shows by how much. Now switch the price index from “number of producers (US, FDA)” to “years since first entry (EU, EC)”: the price in year three is about 11.5 EUR instead of 4.0. The two studies differ by that much because they measure different things in different markets, and your report has to say which one it follows. Then open the sensitivity table: entering one year late hardly changes the present value at your launch (about one per cent, upwards), which hides the delay; the next column, brought to the first generic entry, shows the loss.

### The points are marked, and so are the limits

**Common pitfall. That a curve is a forecast, or that the page tells you when you may sell.** Each curve point says whether a published study states it (*stated*: the FDA's 2019 report for the price by number of producers, the European Commission's sector inquiry fact sheet for the price by time and for the generic volume in the first two years) or the page interpolated, extended or chose it (*assumed*, with its reason); a number you edit becomes *assumed*. The number of rivals and the newcomer's share have no source. The databases that give price-erosion curves by product class are commercial and are not shipped. The page is not a freedom-to-operate analysis: the dossier of the apalutamide case lists the European counterparts of the patent and their supplementary protection certificates as not verified, and the date the page starts from (3 December 2035) is Google Patents' anticipated expiry of one US patent.

### What this tool does not carry

The molecule's volume is held constant and the originator's reply (a price cut, an authorised generic, a new formulation) is not modelled. The sensitivities move one thing at a time on the same curves and carry no probability.

## 5.15 Ways a pharmaceutical-plant economics goes wrong

*Declared anchor:* `ch:economics`.

*Tool:* `tool=economics-errors`. *Case it starts from:* none (one illustrative plant of round numbers); no engine run.

### The question

*My group has written the economics chapter of a plant design. Before we hand it in, which of the mistakes that recur in such chapters are in ours, and which of them would Choupo have caught?*

The page takes eight error classes met in earlier plant-design reports (none is named here, or on the page) and shows, for each, the wrong calculation and the right one on the same editable inputs, side by side. The classes are: a sold quantity that is not the plant's production, $N = \min(Q, Ds)$ against a quantity raised by an unargued factor; a break-even computed on another revenue than the NPV's, tested by putting the break-even back into the NPV; the tax charged on sales instead of on taxable income, $T = t(S - K - d)$; a royalty charged outside the years in which a licensed patent is in force; three working capitals (a fraction of the fixed capital, a fraction of the sales, a build-up from days) that can differ by more than an order of magnitude; a discount rate mistaken for the cause of an internal rate of return, drawn as the NPV against the rate with the IRR where the curve crosses zero; a utility bill out by a unit conversion, with energy and volume refused as a mixture; and a cost per square metre that is a bare number instead of a declaration with a source and a reason.

### The guard under each class

Under every example is the answer to *what does Choupo do about this?*, read from the engine and quoted with the file it was read in, in one of four words: it *refuses* (the run stops), *announces* (the run goes on and prints it), *constructs* (the engine's own calculation is done the right way in one place and published, but nothing compares it with yours), or does *nothing*, and then the page says so in those words. Choupo has no patent calendar (a royalty is charged only inside the window the case declares, and a window with no end is refused), computes the break-even quantity only when the case declares it, does not choose a working-capital method for you (it prints three side by side when asked) and does not judge a discount rate or a unit cost. The test of the page opens the source files it cites and holds each guard sentence to the text it rests on, so a guard that disappears from the engine fails the test instead of staying on the page.

### What to vary, what to watch

**Try it now.** In the sold-quantity example set the factor to one and then to six: the quantity booked goes from the market share to several times the plant's production, and the page marks it impossible. Switch the series to thousands read as millions and watch the same. In the discount-rate example raise the rate to sixty per cent: the NPV of a sound ten-year project turns negative, while the IRR does not move; then multiply the cash flow by six and read an IRR above one hundred per cent that no rate chose.

**Common pitfall. That a green run is a correct economics.** The engine's chain is consistent; it does not know whether the quantity, the price, the royalty or the unit cost you declared are those of the plant you described. Every number on this page is an illustration, none is a market study, and nothing on it is financial or legal advice.

### What this tool does not carry

A year-end discounted cash flow with straight-line depreciation, no inflation, no financing and no construction period; the engine's own appraisal has more than that, and the page does not reproduce it.

## 5.16 What a plant costs to build

*Declared anchor:* `ch:economics`.

*Tool:* `tool=plant-cost`. *Case it starts from:* none (two invented equipment lists, one of them a pharmaceutical tablet plant, and a third list read from a published paper); no engine run.

### The question

*I have a list of equipment. What number do I put on the capital to build the plant, on the money that keeps it running, and on a loan that pays for part of it, and why do the methods disagree?*

The page builds the fixed capital three ways from one equipment list: Lang’s multiplier, $F_{\mathrm{L}} = K\,C_{\mathrm{Be}}$, with his three factors (3.10 for solids, 3.63 for solids with fluids, 4.74 for fluids; H. J. Lang, *Chem. Eng.* **54**(10) 117 (1947) and **55**(6) 112 (1948)); a factored estimate, $D = C_{\mathrm{Be}}(1 + \sum f_i)$ and $F_{\mathrm{f}} = D + D\sum g_j$, in which every fraction is the student’s input, with a pointer (no number) to the book chapter that tabulates it beside it, because a range not checked against its book is not printed; and the bare-module route Choupo itself uses, $1.18\,C_{\mathrm{BM}}$ plus a site factor of 0.50 on the base-case sum. The three are printed side by side with their spread, and the page says that the spread is the lesson. The working capital follows by three methods (itemised in days, a fraction of the sales, a fraction of the fixed capital) with their spread, and a loan is drawn year by year with its interest during construction.

### What the engine says

Every engine claim on the page is quoted with the line it was read at (`EconomicsPass.cpp`, `Turton.cpp`, `CostingModel.H`), and the test of the page opens those files and holds each quoted line to its text.

### What to vary, what to watch

**Try it now.** Switch from the tablet plant to the fluid-processing list and watch Lang’s number move with its factor while the bare-module number moves with the items. Move the piping fraction up and down and read how far the factored estimate moves. In the loan, lengthen the construction by a year and watch the interest during construction grow.

**Common pitfall. That a factor is a quotation.** Every default is an illustration and the page prints no typical range; neither replaces a vendor’s price or your own declared value with its source.

### A worked example from the literature

The third example list is Table 3 of H. Asgharian *et al.*, *Energy Conversion and Management* **315** (2024) 118764 (doi 10.1016/j.enconman.2024.118764), a decentralised green-ammonia plant of about 8.5 kg/h: eight rows from the electrolyser (0.19 M€) to the flash tank (0.015 M€), totalling 0.887 M€. The article states “This is an open access article under the CC BY license”; only the numbers of the table are re-typed, no figure is copied. The authors estimate the cost of establishing the plant with a Lang factor of 4.7 (their reference [50], fluid processing) and print 4.17 M€; the page reproduces it ($4.7 \times 0.88687 = 4.168$, which rounds to 4.17), and shows beside it the 4.74 of Lang’s 1947 paper, which would give 4.20. The paper gives no bare-module factor, so the page sets $F_{\mathrm{BM}} = 1$ as a placeholder and says so.

### What this tool does not carry

No risk analysis, no cost index or location factor, and none of the engine’s per-item material and pressure factors; the equity cash flow and the tax effect of the loan are the engine’s and are not drawn.

## 5.17 What a plant costs to run, and when it pays

*Declared anchor:* `ch:economics`.

*Tool:* `tool=plant-running-cost`. *Case it starts from:* none (one illustrative plant of round numbers); no engine run.

### The question

*I have the mass and energy balance of a plant and an investment. What does it cost to run, at what capacity does it break even, and does it pay?*

The page carries one plant through seven short calculations. The cost of production is itemised in four groups, direct, indirect, fixed and general, with every factor an input and, beside it, a pointer without numbers to the book that tabulates it (Peters, Timmerhaus and West, chapter 6): the student reads it there and declares the value with that source. The raw materials are the balance’s quantity times a price, never a percentage. The same bills are then priced in Choupo’s three terms, $\mathrm{COM}_d = 0.180\,I + 2.73\,C_{OL} + 1.23\,(C_{RM} + C_{UT} + C_{WT})$, with the difference

explained term by term. The crew is derived from the hours a post is staffed and the hours one operator is paid; depreciation is set by asset class against one straight line; the break-even capacity is drawn three ways (accounting, cash, and NPV equal to zero); the cash flow gives NPV, IRR and discounted payback with a discount rate built as $(1 + T_1)(1 + T_2) - 1$; a one-at-a-time sensitivity gives a tornado and the elasticities; and the last step does the market arithmetic (price index, currency, apparent consumption, FOB and CIF, unit margin).

### What Choupo does and does not do

Every engine statement on the page carries the line of `EconomicsPass.cpp` it was read at, and the test of the page opens the file and holds each to its text. Choupo has the three-term cost, a crew from a correlation or a declaration, straight-line depreciation over one life, the cash flow with its NPV, IRR and paybacks, and a one-at-a-time sensitivity, and, when the case declares them, a break-even (the operating one and the NPV one, from the same revenue), a royalty charged inside a declared patent window, a building cost from the rooms' areas priced by grade, and the working capital by three methods side by side. It has no itemised cost of production and no depreciation by asset class.

### What to vary, what to watch

> **Try it now.** Change the shifts per day from three to one and watch the people per post fall and the labour bill with them; then change the maintenance factor and read the NPV move. Put the NPV break-even fraction back into the cash flow and read a net present value of zero. Raise the step of the sensitivity from ten to thirty per cent and see which bar changes length faster than the others.

> **Common pitfall. That a starting value is your plant's value.** The starting values are illustrations, and a one-at-a-time tornado ranks levers; it does not measure risk. The numbers on the page are illustrations, and nothing on it is financial or legal advice.

## 5.18 Where to build it, what to make, and what it sells for

*Declared anchor:* `ch:economics`.

*Tool:* `tool=site-process-price`. *Case it starts from:* none (invented illustrations); no engine run.

### The question

*Before I cost a plant: where does it go, which patented route does it run, and what does the product earn at the plant gate?*

The page does the three pieces of arithmetic that the other economics pages leave out, and ships no figure about any place, patent or market. The *site* is chosen with a weighted matrix: the page lists prompts (proximity to raw materials and customers, utilities, labour, transport, climate, incentives, land), the student declares the criteria, the weights and a score from 0 to 10 for each candidate, and the page computes $S_i = \sum_j w_j s_{ij} / \sum_j w_j$. It then shows how fragile the ranking is: holding the other weights, two sites tie at exactly one weight of a given criterion, $x^* = (R_a - R_b)/(s_{bj} - s_{aj})$, and the page reports, for each criterion, the nearest weight at which the current leader is overtaken, and plots every site's score against that weight.

The *process* is chosen from patents with a method given as a checklist: start from the patents cited in an encyclopedia entry, group them by owner, separate industrial examples from laboratory ones, read the Examples section for temperatures, pressures, times and yields, and check that one patent covers the whole route. A small calculator then turns an example (quantities charged, product isolated, times) into kilograms charged per kilogram of product, the process mass intensity, the E-factor, the yield against the theoretical amount, the cycle time, the product per hour and a scale factor to a target batch. The page cites no patent and invents no number.

The *price* is the margin chain between the shelf and the plant gate. Each stage (a tax, a pharmacy, a wholesaler, a fixed fee) is the student's declared input, in one of four forms that are not interchangeable: a share of the price the stage sells at, a markup on the price it pays, a tax added on top, or a fixed amount per pack. The page ships no margin and no tax rate: the instruction is to read the regulation of the market and declare each stage with its source. The arithmetic runs from the shelf price down to the ex-factory price and back up from the price the plant needs, and the round trip must return the starting price.

### What to vary, what to watch

**Try it now.** Raise the weight of one criterion until the leader changes and read the weight at which it did; compare with the weight the page reported. Swap the fixed fee with the pharmacy and watch the ex-factory price move, then swap the tax with the pharmacy and watch it not move. Change a pharmacy margin from a share of its selling price to a markup on its cost with the same number and read the difference.

**Common pitfall. That the matrix decides, or that a patent example is a design.** The weights and scores are the student’s judgements and the example is one inventor’s experiment; neither is a fact about a site or a plant. Nothing on the page is legal advice, a market study or a freedom-to-operate opinion, and every starting value is an illustration.

## 5.19 How a batch plant is organised: lots, campaigns and the Gantt chart

*Tool:* `tool=campaign-plan`. *Cases it starts from:* `batch/plant/apalutamide01_tablets` and `batch/plant/donepezil02_two_stre` run in the browser by a button on the page; the two toy plants are invented and run no engine.

### The question

*My plant makes several products in lots. Which stage paces it, how many hours of the year does it need, where does a continuous machine such as a spray dryer fit, and does the order of the campaigns matter?*

The page has two halves and says which one you are in. In the **toy** half a plant of five stages (a solution tank, a spray dryer, a blender, a press and a coater) is redrawn as a Gantt chart while you move the lot, the lots per campaign, the changeover, the dryer’s operating time, the demand and the number of tanks; the limiting stage is outlined as it moves, with $T_{\mathrm{L}} = \max_j t_j / N_{\mathrm{op},j}$, each stage’s utilisation and the year’s required hours against the available ones recomputed live. A second toy makes two or three products in single-product campaigns, with a clean at the coating pan that depends on the direction of the change; the order is a drag, and the page lists every order with its hours. In the **engine** half a button runs the case and the page draws the plan Choupo published, with the same bars, and recomputes nothing.

### What the engine says

The toy is a second implementation of `BatchPlanPass.H` (lines 97–123 for one product, 205–224 for several), held by the page’s test to what the engine published for the two cases to $10^{-6}$: every cycle, $T_{\mathrm{L}}$, offset, campaign, busy hour and Gantt bar. With two products the year pays each direction once whichever product goes first (`BatchPlanPass.H`:208–210), so donepezil02’s two trains need 880.2 h and 2079 h in either order; from three products the order matters, and only through the directional cleans (`BatchPlanPass.H`:219–220). A semicontinuous stage that the case rates (apalutamide01’s spray dryer, `operatingTime fromFlowsheet`, postDict line 178) runs through the lot, the tank that feeds it is held for its whole run (`dischargeTime throughFedStage`, postDict line 170), and the dryer is cleaned at the campaign changeover, not between lots.

### What to vary, what to watch

**Try it now.** With one solution tank the tank is outlined as the limiting stage: it is held through the dryer’s run. Choose two tanks and the outline jumps to the dryer. Then lengthen the dryer’s operating time: the tank’s cycle grows with it, and the page warns that the plan asks the dryer for a rate it was not rated at. In the second toy, make the third product and reorder the campaigns: the hours differ between the two cycles, by exactly the difference of the directional cleans.

**Common pitfall. That the cheapest order is the right one.** It is the order with the fewest hours; the demand, the shelf life and the safety stock decide the rest. The engine takes the order as given and never optimises it.

### What this tool does not carry

One train and no intermediate stores (a real plant decouples trains with stores, each with its own hours), no mixed sequencing inside a campaign, no per-product mass balances, and a clean is hours here, not a cleaning validation (EU GMP Annex 15).

## 5.20 How tall must the stack be? Gaussian plume dispersion

*Declared anchor:* `ch:stack-dispersion`.

*Tool:* `tool=stack-dispersion`. *Case it starts from:* `steady/gibbs/gibbs06_thermal_nox`, run in the browser by a button on the page, whose outlet stream `flueGas` is read; the toy half runs no engine.

### The question

*How tall must my stack be so that the gas it emits reaches the ground diluted enough? Why does a taller stack lower the ground concentration, why is a calm clear night the bad case far from an elevated source and a sunny afternoon the bad case close in, and how does a limit value become a stack height?*

The page has two halves and says which one you are in. In the **toy** half an emission rate you set feeds four live figures: Turner's key to the stability classes with the cell for your weather lit (day or night, sun or cloud, wind), whose class colours the other figures; the plume in side view with its $\pm\sigma_z$ envelope and the image source below the ground, with sliders for the stack height, the wind, the exit velocity, the diameter and the temperatures; the ground-level concentration along the wind on log axes, the six classes drawn faint, the maximum marked with its distance and your limit as a horizontal line; and a plan view of ground isopleths with a fence of the radius you choose, so that the maximum is seen inside or outside it. The page then inverts the question: the least stack height for which the worst of the allowed classes keeps the maximum under your limit, beside the good-engineering-practice height against building downwash. In the **engine** half a button runs the methane burner, reads the `flueGas` stream (its flow, temperature, pressure and the mole fraction of NO) and converts it to an emission rate and a buoyancy flux for a diameter and a stack-gas temperature you set.

### What the engine says

Nothing about dispersion: Choupo has no dispersion model, no fired-unit type and no stack as a sized item (its combustors are Gibbs reactors). The stream is read and never recomputed; $Q$ is the stream's NO flow times the molar mass from the catalogue, held by the page's test to the flow the engine published. The dispersion is the page's own arithmetic, held by hand calculations written in its test: the image-source slope at the ground, the normalisation of the plume, a worked example, the closed-form maximum $2Q/(\pi e u H^2)\,\sigma_z/\sigma_y$ at $\sigma_z = H/\sqrt{2}$, and the inversion (the reported height puts the maximum at the limit to one per cent). *Which coefficients were read:* the urban sigma set and the good-engineering-practice formula; the open-country set, Turner's table of classes and Briggs' rise constants were written from the standard tabulation and are marked unread on the page.

### What to vary, what to watch

**Try it now.** Choose a sunny afternoon at 2.5 m/s and read the ground-concentration figure: the unstable curve is high and close to the stack. Switch to night, clear: the class becomes F, and far down the wind its curve rises above the unstable ones while its maximum slides to the edge of the window. Raise the stack height and watch the maximum fall roughly as $1/H^2$; then type a limit, its source and its date, and read the least stack height, and move the fence to see whether the maximum falls inside it.

**Common pitfall. That the stack height the page gives is a permit.** It is the answer of a lesson's plume model to a limit you typed. The limit is yours, with its averaging time; an air-quality assessment follows the model its regulator names.

### What this tool does not carry

Flat ground, no terrain and no building wake beyond the good-engineering-practice rule; no chemistry, deposition or washout (the NO of a flue gas turns into $NO_2$ in the air); the 10 m wind at every height and the whole plume rise taken at the stack; no averaging-time correction; a cold jet's momentum rise not built; maxima outside 100 m to 10 km only flagged.

## 5.21 Demand forecast for a drug product

*Declared anchor:* `ch:economics`.

*Tool:* `tool=demand-forecast`. *Case it starts from:* none (an invented illustrative history); no engine run.

## The question

*What yearly volume should our plant's economics read, how sure are we, and how much of it is the market's and how much is our assumption?*

Three methods run on one editable history of the market, in packs or in defined daily doses. $A$, a straight line fitted by least squares, with the slope, the intercept, $R^2$, the standard error, the $t$ statistic and $p$ value of the slope and the prediction interval of a future year $\hat{y}_0 \pm t_\nu \, s\sqrt{1 + 1/n + (t_0 - \bar{t})^2/S_{tt}}$. $B$, a dose count $P = N\,p\,q\,m\,c\,u\,365/D_{\text{pack}}$ from the population, the prevalence, the treated fraction, the treatment share, the adherence and the doses in a pack, reconciled against the last observed year. $C$, the market volume times a logistic share $S(j) = S_{\max}/(1 + e^{-k(j - j_{1/2})})$.

## Units, and the capacity factor

The page asks the unit of the series and of the answer (units, thousands, millions) and computes nothing until both are said; it refuses a series whose adjacent years differ by a factor of ten or more, which is how thousands pasted among millions looks, unless the jump is confirmed; and it warns when the plant and the forecast are two orders of magnitude apart. What the plant sells is the market volume times the share it holds times a *capacity factor*. The factor is shown as a line, together with the factor that would fill the plant and the sold quantity as a fraction of the whole market; a year in which the plant would have to sell more than the whole forecast market is marked. The factor is an assumption about the world, not a property of the plant.

## What the case reads

The output is the pair the batch plan's economics reads (`src/postProcessing/EconomicsPass.cpp`): `rampUp`, the fraction of the design production sold in each operating year with each entry in $[0, 1]$ (a year the market would exceed the plant is capped at one and marked plant-limited), the design production in kilograms through the tablets in a pack and the mass of a packaged tablet, and `streamFactor fromBatchPlan`. The price by year is not here; it is the subject of the patent-expiry tool.

> **Try it now.** Fit the default history and read $R^2$ and the band: a good fit to ten years says the line describes them, and the band says how far a new year can sit from it. Change the capacity factor from one to six and read the sold quantity as a fraction of the market. Then change one number of the series by a factor of a thousand and watch the page refuse.

> **Common pitfall. That a forecast is a measurement.** The trend assumes the cause of the past continues, the dose count assumes five factors you estimated, and the S-curve assumes a ceiling and a speed; none of the three is tested by any statistic on the page. The three disagreeing is the finding.

## What this tool does not carry

Only the volume is forecast, not the price; the history on the page is invented and must be replaced by a series you can cite.

# 6 The shelf "Property data & parameters"

Where a property number comes from, whether it may be published, when a record can be believed, and how a measurement becomes a fitted parameter: none of it is a thermodynamic *model*, though every page uses one, so these pages have a shelf of their own, right after Thermodynamics. The path *From a measurement to a parameter* walks them in order: *Where do properties come from?*, *Where to find property data*, *When a property database lies to you*, the local-composition models, *least squares*, *From a database to binary parameters*, *a pair from solubility data* (`sle-pair`), *salt mixing* (`salt-mixing`) and *Four ways to price a mixture*.

## Where to find property data (`tool=property-data-sources`)

Which property, which system and which conditions pick the kind of source; a map of free sources with what each holds, how to cite it and what its licence allows; the classes of source that cannot enter a public project; measured, fitted and estimated values; and a real catalogue record read line by line. The page computes nothing: its numbers are the catalogue's own. *Declared anchor:* `ch:criticals`.

## From a database to binary parameters (`tool=database-to-parameters`)

Choupo's curation pipeline as a lesson: a ThermoML file is read in the browser (the same reader the curation tool is held to), the usable blocks are chosen, the NRTL pair is fitted by the engine's own rule, and the fit is graded with the cap a single laboratory or temperature cannot lift. *Declared anchor:* `ch:criticals`.

## A pair from solubility data (`tool=sle-pair`)

The activity coefficient of a solute is extracted from measured solid solubility (benzoic acid in four solvents), an NRTL pair is fitted with every fourth point held out, and the page shows that a solubility-only pair fixes the solvent-rich end of the composition range and not the other. *Declared anchor:* `ch:fittable`.

## Pitzer mixing parameters from a solubility curve (`tool=salt-mixing`)

The mixing parameters $\theta_{\mathrm{Na,K}}$ and $\psi_{\mathrm{Na,K,Cl}}$ are fitted to the measured NaCl–KCl–$H_2O$ solubility branches at one temperature and tested on the temperatures held out. *Declared anchor:* `ch:pitzer-hmw`.

## 6.1 Choosing an electrolyte activity model

*Tool:* `tool=electrolyte-models`. *Case it starts from:* none in the corpus — the page writes its own props cases and runs them with `choupoProps` in the browser.

### The question

*I have a solution with salts; which activity model should I declare, and what do I lose with the others?*

The page starts from the term every model shares, the Debye–Hückel limit $\ln\gamma_\pm = -3A_\phi|z_+z_-|\sqrt{I}$ with the engine's $A_\phi = 0.3915$ at 25 °C, drawn alone against the measured table of the salt the student picks. It then writes what Davies (a fixed correction, charge only), Pitzer (short-range virial terms per cation–anion pair, with a $\beta^{(2)}$ term in the records that carry one) and eNRTL (a local-composition term with water and the salt as species) add, each equation read from the source file it names. The student picks NaCl, KCl, $CaCl_2$ or $Na_2SO_4$; the page draws $\gamma_\pm$ and $\phi$ under each model against the critically evaluated values of Hamer and Wu (1972), Goldberg and Nuttall (1978) and Goldberg (1981) that the tree carries in `data/curation/electrolytes/`, tabulates the mean and worst deviation, turns the water activity of each model into the osmotic pressure a membrane must overcome, asks four questions (dissolved acid gases, a second solvent, a second salt, the ionic strength) whose answers name a model, and ends with $MgSO_4$, for which the catalogue holds a Pitzer record and no eNRTL record.

### What the engine says, and what the page computes

The engine computes every model value. Pitzer and eNRTL run through the package a process unit declares (`formulation electrolyteGammaPhi` with `activityModel { model Pitzer; }` or `model eNRTL;`) and the props operation `electrolyteActivity`, one operation per molality; Davies runs through the speciation solver (`speciate`, `activityModel davies`), once with the network restricted to water's dissociation and once with every equilibrium the catalogue holds for the ions — for $Na_2SO_4$ that forms $NaSO_4^-$. Every model is asked at every measured molality, so no deviation is interpolated. The page computes, and prints with its constants: the limiting law, the stoichiometric mean $\gamma_\pm$ of the run with ion pairs from the free-ion activities the engine wrote, Davies' $\phi$ from the engine's $a_w$, the measured $a_w$ from the table's $\phi$, the osmotic pressure $\Pi = -(RT/\bar{V}_w)\ln a_w$ with the engine's $\bar{V}_w = 1.807 \times 10^{-5}\,\mathrm{m^3/mol}$, and the deviations. Whether a model was regressed to the table on screen is read from the record's own `source` and `fittedInCase` lines: the eNRTL records of KCl, $CaCl_2$ and $Na_2SO_4$ were, so those comparisons are marked in-sample, and for $CaCl_2$ and $Na_2SO_4$ the rows the regression held out are drawn

open and scored apart. The $MgSO_4$ eNRTL run is shown as the engine's refusal, verbatim. The design record `docs/design/choosing-an-electrolyte-model.md` holds the measured numbers for each salt.

### What to vary, what to watch

**Try it now.** On NaCl, read where Davies leaves the table: within 0.3 % at $I = 0.1$ mol/kg and about 8 % at $I = 0.5$. Switch to $Na_2SO_4$: Davies on free ions is 21 % off already at the first row ($I = 0.3$), and the curve with the catalogue's $NaSO_4^-$ ion pair comes back to within 3 % up to $I = 0.6$ mol/kg — the specific interaction Davies lacks, declared as chemistry — and leaves again above it (+19 % at $I = 1.8$). On $CaCl_2$, compare Pitzer and eNRTL: eNRTL was fitted to this very table and still misses it by up to 4.6 %. In step 4, pick 1 mol/kg of NaCl: Davies' water is ideal, so its osmotic pressure is the ideal one, van 't Hoff's law in its molality form (pure-water molar volume, not the solution's molarity).

**Common pitfall. A small error on a fitted table.** An error on the points a model was regressed to shows that the fit worked, not that the model predicts; the page says which comparisons are in-sample. The Pitzer records and the NaCl eNRTL record are not regressed to a table in this tree, but their authors fitted them to measurements of the same salts at 25 °C, so their small errors there also show the quality of a fit, not a prediction.

### What this tool does not carry

Everything is at 25 °C, with no temperature control: 96 of the catalogue's 99 Pitzer pair records hold their parameters at 25 °C only, and the single-salt eNRTL scales its two energies by $298.15/T$ and nothing else (the engine's own simplification, not the form of Chen and Evans, 1986, eq. 45); neither is tested here away from 25 °C. The measured tables start at 0.1 mol/kg. Only single salts in water are compared with measurements; `pitzerHMW`, the mixed-solvent eNRTL and `edwardsPitzer` appear only as answers in step 5. Davies is drawn up to $I = 2$ mol/kg. The osmotic pressure is the thermodynamic minimum, with the pure-water molar volume and the bulk composition. *Declared anchor:* `ch:electrolytes`.

## 6.2 Residue curve maps: azeotropic and extractive distillation

*Tool:* `tool=residue-curves`. *Case it starts from:* none in the corpus — the page writes a props case of its own and runs it with `choupoProps` in the browser.

### The question

*Which products can a simple column make from a three-component feed that has azeotropes, and why does adding a heavy solvent, changing the pressure or splitting the liquid let a column do what it otherwise cannot?*

The page starts from the batch still of the Rayleigh page with a third component: the pot liquid of a simple still follows a residue curve, $\mathrm{d}x_i/\mathrm{d}\xi = x_i - y_i^*(x)$ with $\xi = \ln(W_0/W)$, and its bubble temperature rises along it. A preset (ethanol + water + ethylene glycol, isopropanol + methanol + water, chloroform + methanol + methyl ethyl ketone, benzene + methanol + methyl acetate, acetonitrile + ethanol + water), a pressure and a grid are chosen; the page draws the map with arrows towards rising temperature, the singular points (pure components and azeotropes) as unstable nodes, stable nodes and saddles in a table with their temperatures and compositions, the distillation boundaries out of the saddles, and the regions they separate. A click places a feed: its own residue curve, its region, and the direct and indirect limiting splits on a straight mass-balance line, with $D/F$ from the lever rule. The page ends with the ways round a boundary (a heavy entrainer that adds none, a heterogeneous entrainer and a decanter, a pressure change, a recycle) and with published azeotropes beside the engine's.

### What the engine says, and what the page computes

The engine computes every equilibrium number: one `propertyPoint` per composition of a triangular grid asks for `T_bubble` and `y_eq_<component>` at the chosen pressure, with NRTL for the liquid (each pair resolved from the catalogue) and an ideal-gas vapour. The pair table shows where the engine found each pair, its fit grade, and every sentence the engine said about it, including each use outside its validity span. A second run asks again at finer compositions around each azeotrope the page located, and runs the engine's liquid-liquid flash (`propertyScanTernary`, mode `lle`) over the triangle at the lowest bubble temperature of the map.

The page computes the rest, and says so beside each figure: linear interpolation inside each grid cell, fourth-order Runge–Kutta on the normalised direction of $x - y^*$, the singular points (located to one grid cell, or to a tenth of one on an edge and a quarter inside after the refinement), their type read twice — from the boiling temperatures around

them and from the eigenvalues of $\partial(x - y^*)/\partial x$ — the boundaries, the regions and the lever rule. It reports its own checks: how many curves start and end at a node, whether any two drawn curves cross, and the largest fall in temperature along a curve that the interpolation leaves. The design record `docs/design/residue-curve-maps.md` holds the arithmetic, the checks measured on every preset, the reading of the published azeotropes, and a proposal for an engine operation that would make the page a pure drawing.

## What to vary, what to watch

**Try it now.** Open the glycol preset: one region, the ethanol–water azeotrope the unstable node and the glycol the stable node — every residue curve runs from the azeotrope towards the glycol: no distillation boundary blocks the split. That the glycol then separates the water in an extractive column is an effect of relative volatility, which the entrainer raises; the map shows only that no boundary stands in the way. Switch to isopropanol + methanol + water and place a feed on each side of the red boundary: the region, and the bottoms the direct split can reach, change. On chloroform + methanol + methyl ethyl ketone, change the pressure from 1.01325 to 5 bar and watch an azeotrope disappear from the model’s map. On acetonitrile + ethanol + water, compare the two grids: the model’s ternary azeotrope lies a few hundredths of a kelvin below a binary one, and the coarse grid does not agree with itself about its type.

**Common pitfall. That a boundary on the map is a boundary in the plant.** The map is the model’s: many catalogue pairs are regressions to one laboratory over a narrow range, used here outside it, and the page prints the engine’s own sentences about each. An azeotrope that appears or vanishes on the map may not do so in the mixture.

## What this tool does not carry

No liquid–liquid split is drawn: every bubble point is for one liquid, and the split is checked at one temperature only. No column is solved; the splits drawn are the limits the rule of thumb allows, and a curved boundary can be crossed by a straight mass-balance line. Residue curves stand in for column profiles, which they approximate at total reflux. The vapour is ideal and the vapour pressures are the catalogue’s correlations, extrapolated at the higher pressures, as the engine announces. Measured azeotropes are compared for four binaries only, read from measured points the corpus carries.

# 7 Constructions not yet documented here

The tools written up above are not the whole registry: several live tools are *not* written up in this guide. The six below are listed with what exists and what is missing, because an admitted gap is a better guide than six thin sections; the remaining ones — the theory lessons reached from the Methods workspace, and the constructions added since this section was written — carry their lesson text in the app, and the registry (`gui/src/ui/methods/registry.ts`) is the authority on the complete list. Each is a working tool: what is absent here is the lesson material, not the construction.

For every entry below, what is missing is the same three things:

- the **misconception** the construction exposes — the part that has to be observed in a classroom before it can be written down, and the part that cannot be guessed from the source;
- **what the picture does not say** — the limits, stated with the same care as the six above;
- a **worked "what to vary, what to watch"** sequence, tried on the tool rather than inferred from its knobs.

## Choosing a solvent, or a solvent mixture (`tool=solvent-mixture`)

Which solvent, or which blend of two, for a given substance: Hansen's three parameters with the temperature they were placed at, the distance of a solvent from the solute's sphere and the relative energy difference where a radius was measured, the volume-fraction mixture rule as a checked rule of thumb, and a phase check that *refuses* to average a pair the activity model finds two-phase. The student predicts the best composition before the engine's curve is shown (a synthetic pair whose minimum sits at 50:50 by construction), then meets Hansen's own polystyrene with hexane + acetone and hexane + nitromethane beside the dissolution ratings he recorded. Every number is the `hansenMixture` operation's, read from the three witnesses under `tutorials/props/solvents/`; the theory and the approximations are in the Theory Guide's section on $R_a$ and RED. *Declared anchor:* `ch:hansen-mixture`.

## Psychrometric chart (`tool=psychro`)

The humid-gas state map: saturation, relative-humidity lines, adiabatic-saturation and true wet-bulb lines, over any carrier gas and condensable pair the catalogue can serve. Every line is engine-computed. *Declared anchor:* `ch:drying`.

## Drying curve, batch tray (`tool=drying`)

The two classical drying plots side by side on one run — moisture against time, and rate against moisture — with the critical moisture visible as the corner where a flat rate begins to fall. The tool is careful about one thing that deserves writing up properly: the engine omits the break time entirely when the crossing was never observed, and the tool reports *which* absence it is rather than drawing a zero. *Declared anchor:* `ch:drying`.

## Cooling tower, Merkel (`tool=merkel`)

The saturated-air enthalpy curve above, the operating line below, and the gap between them shaded — that gap is the driving force the packing has to buy. A chip holds the four-point Chebyshev hand method against the engine's converged integral. *Declared anchor:* `ch:coolingTower`.

## Batch still, Rayleigh (`tool=rayleigh`)

The graphical Rayleigh integration: the area under $1/(y^* - x)$ between the pot's final composition and the charge *is* $\ln(W_0/W)$, drawn over the engine's own equilibrium curve and judged against the engine's rigorously integrated still. *Declared anchor:* `ch:rayleigh`.

## Pump against system curve (`tool=pump-system`)

The operating point as an intersection: the pump's pressure rise falling with flow against the pipe system's demand rising with it. The tool is explicit that the "pump curve" is whatever the pump *model* published, and greys it out with the caveat spelled in full when the model was specified by a fixed pressure rise and the curve is therefore the specification echoed back. *Declared anchor:* `ch:rotating`.

## Adsorption breakthrough (`tool=breakthrough`)

The S-shaped breakthrough curve against the ideal square wave drawn at the engine's stoichiometric time: the mass-transfer zone consumes bed capacity long before the bed is saturated. The witness runs the engine's default second-order van Leer scheme ($t_b(5\,\%) = 2934$ s, a 231.5 s wide front, 1.75 cm unused, within 0.18 % and 5 % of the mesh-converged front); declaring `convectionScheme upwind;` shows the first-order grid widen the front to 313.5 s. *Declared anchor:* `ch:adsorption`.

# 8 Using a tool in a lesson

Three patterns that work, drawn from how the tools are built rather than from how they could be used.

## Turn one knob, and predict first

Every classroom tool re-solves in the browser on each knob change, which makes the interesting move cheap: ask for a prediction, then turn the knob. The productive knobs are the ones whose consequence is counter-intuitive — $\Delta T_{\min}$ on the pinch tool, area on the $\varepsilon$–NTU chart, `T_guess` on the Van Heerden diagram, the solute loading on the Kremser tool. All four are documented above with what should happen.

## Make the deviation chip the discussion

Several tools put a classical hand method beside the engine's answer and report the gap. That gap is the lesson, and its size is not the point — its *cause* is. On the Kremser tool the gap is the isothermal assumption failing, and it grows the more exothermic the duty. On the cooling-tower tool the gap is nearly zero, and saying so plainly rather than manufacturing drama is itself worth modelling for students: a hand method that reproduces a converged integral to five figures has been *validated*, and that is a result.

## Read the refusal

An engine that refuses is teaching. A pinched McCabe–Thiele construction, an infeasible cooling tower, an absorber whose stripper twin does not activate the tool, a drying run that never reached its critical moisture — each of those is a message with a reason in it. The tools show the engine's own words verbatim rather than a generic failure, and the reason is usually the physics of the case rather than a mistake in the input.

**Key idea.** All models are wrong; some are useful. Everything in this plane is built so a student can see *which* assumption is doing the work — and what happens on the day it stops being true.

# 9 Notes for maintainers

## One anchor, and it must resolve

Each tool's registry entry names a Theory Guide destination. The preamble shared by all the manuals sets `destlabel=true`, so every `\label` in the Theory Guide becomes a PDF named destination and the book icon jumps straight to it. An anchor the guide does not define fails *silently* in a browser — the PDF simply opens at page one — so a test in the application checks every declared anchor against the guide source. A tool without a resolvable anchor is not finished.

## Anchors are chapter-level, and some chapters are broader than their tool

The declared anchor points at the section that owns the tool's *unit operation*, which is not always the section that derives the tool's *construction*. Where the two differ, the *Where the derivation lives* paragraphs above name both destinations rather than pretending the first covers everything. Two cases are worth knowing about before adding a seventh tool section:

- A construction may be the hand method for a chapter that derives the *rigorous* model instead — the staircase against the stagewise column, the Levenspiel areas against the differential balance.
- A construction may be argued most fully in its *witness case's* own dictionary header rather than in the manual. That is a real source and this guide cites it as one, but it is not a substitute for a derivation, and a section that has to reach for it is flagging a gap in the Theory Guide rather than filling one.

## When a tool changes

A knob added or removed, a chip relabelled, a witness re-parameterised: each of those can invalidate a *What to vary, what to watch* block here, and none of them will fail a build. The blocks are written to name knobs by their on-screen labels so a mismatch is at least visible to a reader with the tool open. Prose staleness is deliberately not gated in this project — a text search cannot tell a live claim from a recorded one — so the check is human, and it belongs in the same change that touches the tool.